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Additive manufacturing of ceramic-metal multi-material structures: technologies and applications



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Abstract

The integration of ceramics and metals, two materials with inherently contrasting properties, presents an attractive and valuable proposition. Additive manufacturing significantly amplifies the ability to create ceramic-metal multi-material structures (CMMMS) featuring complex and nested architectures. This review provides an in-depth summary of the cutting-edge technologies for additive manufacturing of CMMMS, outlining their traits, capabilities, limitations, and future trajectories. Further, we emphasize the unique functional benefits that additive manufacturing offers for applications—fabrication of integrated functional systems and customizable multi-functional integration and—summarize methods for embedding multiple devices within printing processes. Finally, we outline key future directions in this field. This review aims to inform and inspire further exploration in this burgeoning area.



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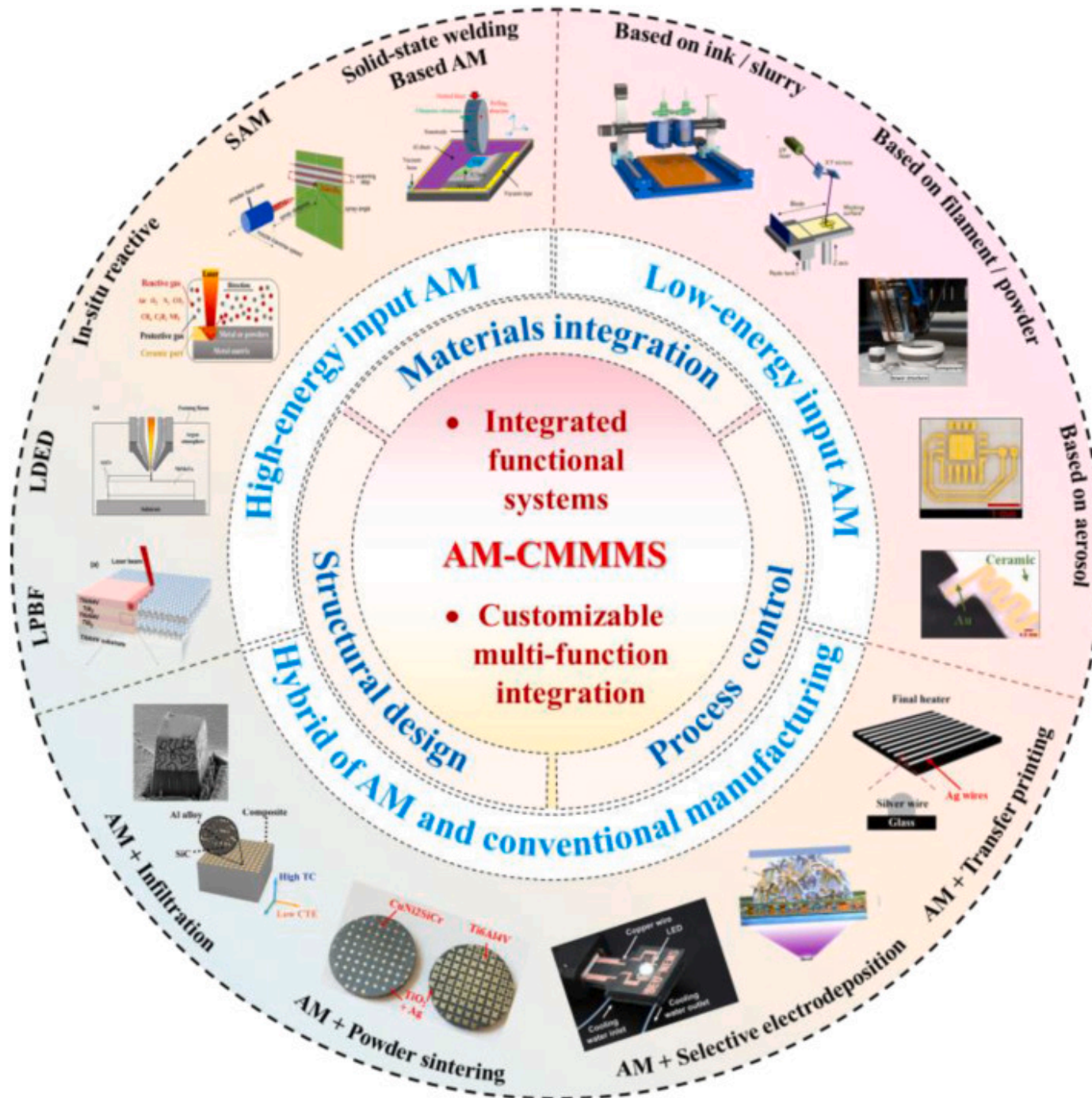
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Keywords

Additive manufacturing; 3D printing; Multi-material structures; Ceramic; Metal

1. Introduction

Metals and ceramics are among the three major categories of engineering materials in modern society, alongside polymers. Metal materials exhibit excellent toughness, good electrical and heat transfer properties, although they suffer from poor corrosion resistance, susceptibility to oxidation, and low high-temperature strength. On the other hand, ceramic materials possess exceptional hardness, strength, wear resistance, corrosion resistance and high-temperature tolerance. Functional ceramics, in particular, demonstrate a wide range of properties such as various electromagnetic properties, insulation capacity and biocompatibility. However, the brittleness of ceramic materials leads to issues with reliability and susceptibility to damage. Consequently, the development of ceramic-metal multi-material structures (CMMMS) has been pursued with the aim of achieving unique combinations of antagonistic properties, such as toughness/strength, conductive/dielectric and magnetic/non-magnetic [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#)], as shown in [Fig. 1](#). CMMMS can also integrate sensor or actuator functions, potentially leading to the development of integrated systems such as microrobots with active sense and move functions [\[9,10\]](#).

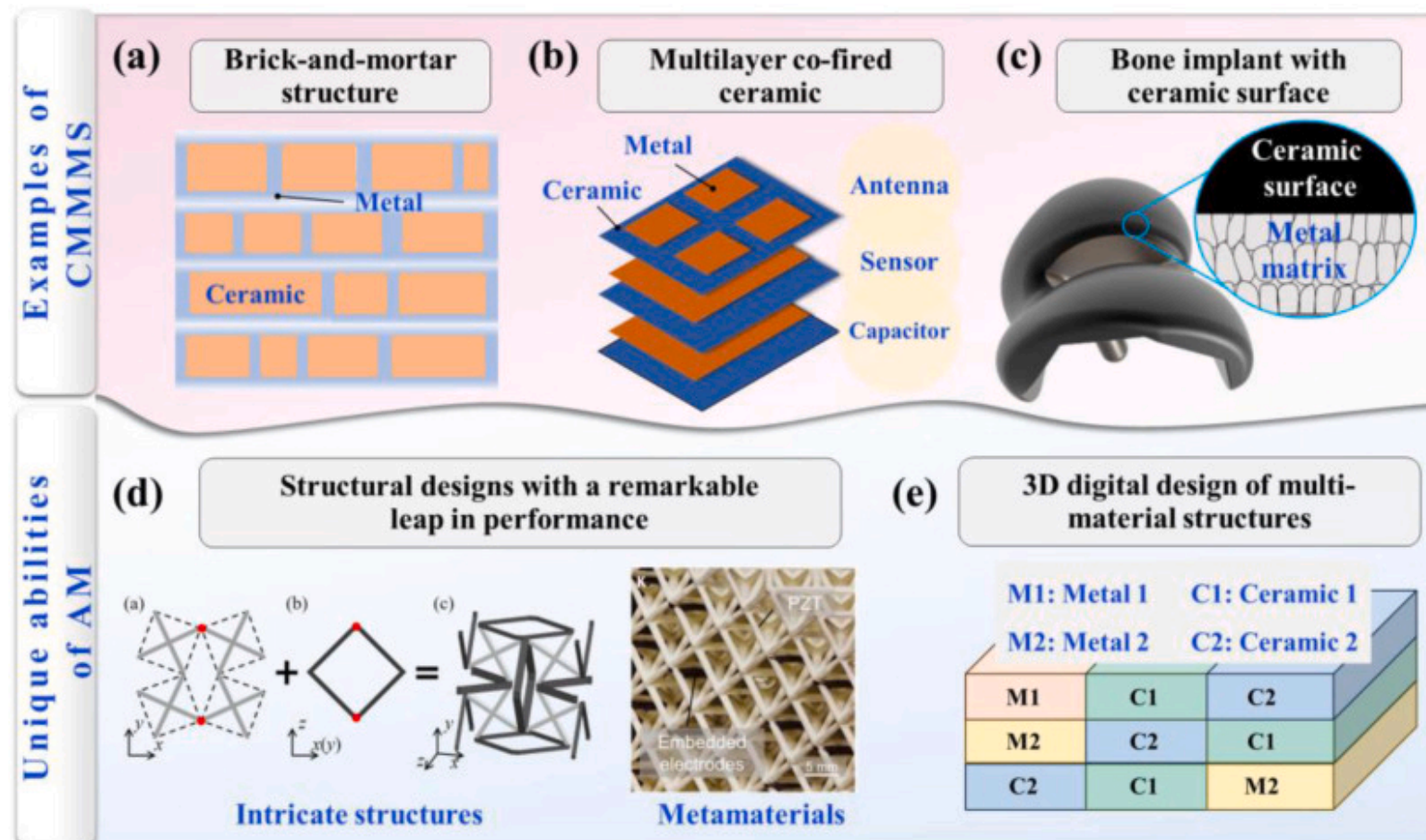


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Fig. 1. An overview of additively manufactured CMMMS and various technologies. Images source from Refs. [2,9,[20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31]].

Therefore, CMMMS are anticipated to overcome the limitations of monolithic materials [11], offering promising applications in various fields, including mechanical structural components [12], electronic devices [13], energy facilities [14] and biomedical applications [15]. Drawing inspiration from the natural nacre structure, CMMMS exemplify the achievement of an excellent balance between antagonistic mechanical properties. Nacre, composed of aragonite platelets interspersed with biopolymers, showcases a combination of high strength and high toughness. Metallic phases can be used to provide toughness, replacing biopolymers [[16], [17], [18]]. As a result, “brick-and-mortar” structures (Fig. 2(a)) derived from nacre can be utilized to address the limitations of ceramics and metals. In the field of electronic devices, high-performance devices also heavily rely on CMMMS. Multi-layer co-fired ceramics (Fig. 2(b)) are a typical example, where the distinct electromagnetic properties of ceramics and metals are utilized. In the biomedical field, metal implants with a ceramic surface (Fig. 2(c)) possess both mechanical properties and excellent biocompatibility, which is of great benefit to bone regeneration [19]. In summary, the fabrication of CMMMS is particularly enticing due to its potential to achieve an array of unique functionalities.



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Fig. 2. Typical application scenarios of CMMMS and the unique advantages of AM: (a) Nacre-inspired “brick-and-mortar” structure, (b) Multilayer co-fired ceramic structure applied to electronic devices, (c) Schematic diagram of metal bone implant with ceramic surface, (d) Unique structural designs [10,34]. (e) 3D digital customization of CMMMS.

Notably, the incorporation of special structural designs marks a significant advancement in material performance, potentially unlocking revolutionary functions. For instance, kirigami structures have dramatically increased the ultimate strain of sheets, boosting it from 4% to 370% [32]. Intricate structures and metamaterials have each demonstrated remarkable leaps in

performance [[33], [34], [35]]. However, traditional manufacturing methods such as pressing, casting, and machining often fall short when it comes to producing complex structures, especially with ceramic materials [36]. Fortunately, additive manufacturing (AM), also known as 3D printing, is emerging as a viable solution to address these limitations, as shown in Fig. 2(d). Additionally, traditional technologies such as brazing and lamination [37,38] become inadequate when dealing with hybrid nested multi-material structures. Moreover, as another unique advantage, the developed multi-material AM technologies have demonstrated great potential in producing multi-material structures [[39], [40], [41]]. Ceramics and metals can be integrated into a component during manufacturing process. By precisely controlling the distribution of materials in 3D space, AM enables the 3D digital design of multi-material structures (Fig. 2(e)), leading to significantly enhanced performance [42,43]. Furthermore, the integration of multiple components or even devices can be achieved. In summary, by leveraging the distinct properties of ceramics and metals, along with the unique advantages of AM, it is expected to yield CMMMS with innovative functions.

Currently, numerous technologies have been proposed for the preparation of CMMMS utilizing AM (as shown in Fig. 1). A comprehensive review of these technologies is essential. However, to our knowledge, there is a paucity of reviews specifically addressing additive manufacturing ceramic-metal multi-material structures (AM-CMMMS). While several notable reviews [40, [44], [45], [46], [47], [48]] have touched upon multi-material AM, they have not provided a thorough introduction or in-depth discussion on AM-CMMMS. Besides, Refs. [13,49] focused on 3D printed multi-material structures for electronic applications, while Ref. [50] concentrated on graded composite materials. Given the significance of AM-CMMMS, this work conducts a comprehensive review on current technologies and their comparison, applications, as well as unique advantages for AM-CMMMS.

2. Technologies

2.1. An overview of AM technologies for CMMMS

AM produces final parts by continuously adding material to the substrate or formed parts. Parts are designed using computer-aided design software and converted into a specific slicing format. Subsequently, the printer adds specified material to designated location based on the slice file. Currently, AM technologies for ceramics and metals have been extensively studied, moving beyond their initial development stage. Several excellent reviews have provided a comprehensive summary of AM technologies for ceramics and metals [[51], [52], [53], [54], [55]].

According to the ISO/ASTM 529000:2021 standard, AM technologies are categorized into seven groups: powder bed fusion (PBF), directed energy deposition (DED), material jetting (MJ), vat photopolymerization (VP), material extrusion (ME), binder jetting (BJ), and sheet lamination. Each category can be further subdivided into various specific techniques. [Table S1](#) presents the process flow and features of commonly used AM technologies for AM-CMMMS.

These technologies are classified basing on two primary classification methods, which align with the classification ideas employed in the subsequent introduction of AM technologies for CMMMS. The first method categorizes technologies into high-energy and low-energy input AM technologies. The former employs high-energy sources like lasers or electron beams to melt or sinter materials, creating robust parts with final functions. In contrast, low-energy input AM technologies involve a relatively mild printing process, resulting in a loose and fragile green part that requires post-treatment, such as debinding and sintering, to remove binders or resins and achieve densification. The second classification method is based on the state of the printing feedstock: solid (powder or wire), liquid (slurry or ink) or aerosol. Researchers can use these classifications to select appropriate technologies for AM-CMMMS. This review focuses on AM technologies for fabricating bulk CMMMS. Techniques such as coating and surface functionalization are discussed as references, but are not in the main scope of this paper.

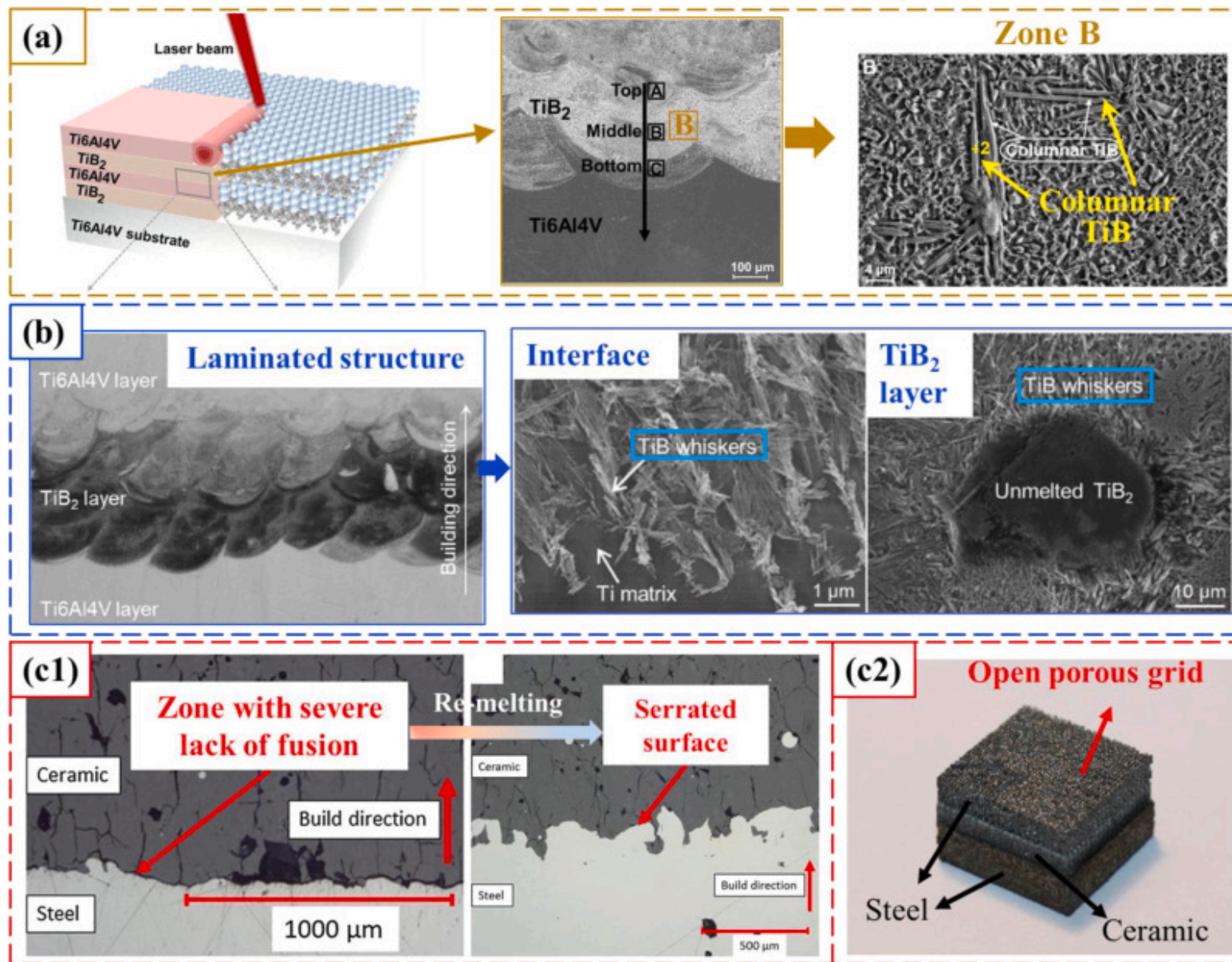
2.2. High-energy input AM

2.2.1. Multi-material LPBF

Laser powder bed fusion (LPBF) works by spreading a thin layer of metal or ceramic powder onto a build platform, then using a laser to fuse the powder according to the cross-sectional contours of a 3D model. After each layer is fused, the platform lowers, a new layer of powder is spread, and the process repeats until the object is complete. During the process, laser serves as a high-energy source.

Previous works primarily focused on enhancing ceramic-metal interfacial bonding quality and yielding substantial improvements in mechanical performance. In the context of bonding dissimilar materials like TiB_2 and Ti6Al4V , Wang et al. [20] discovered that the formation of TiB generated by the reaction of TiB_2 and Ti6Al4V at their interface served as a transition phase that significantly improved bonding, as shown in [Fig. 3\(a\)](#). Further optimizing the multi-material LPBF process parameters for these materials they found that using higher laser power and lower scanning speeds effectively melted the ceramic powder without inducing defects like transverse cracking or shrinkage [56]. Building on this, the researchers created a bionic, layered TiB_2 – Ti6Al4V structure with

alternating soft and hard layers [57], as shown in Fig. 3(b). The ceramic TiB_2 layers required a higher laser power and faster scanning speed (350–450W, 200–1000mm/s) than the Ti6Al4V layers (250W, 600mm/s). Notably, the interface revealed in-situ growth of TiB nanocrystals, a result of the TiB_2 particles reacting with the metal matrix, facilitated by Marangoni effect and the high temperature gradient at the interface. The Marangoni effect is generated by the substantial temperature gradient in the molten pool. The rapid heating and cooling during the printing process create a steep temperature gradient across the molten pool [58], decreasing from the center of the free surface toward the periphery. This temperature gradient induces a significant surface tension gradient, driving fluid flow from the interior to the outer edges of the pool and forming Marangoni convection vortices. Meanwhile, previously deposited material that is remelted due to heat conduction is transported into the molten pool and thoroughly mixed, thereby greatly enhancing the uniformity of elemental distribution in the pool.



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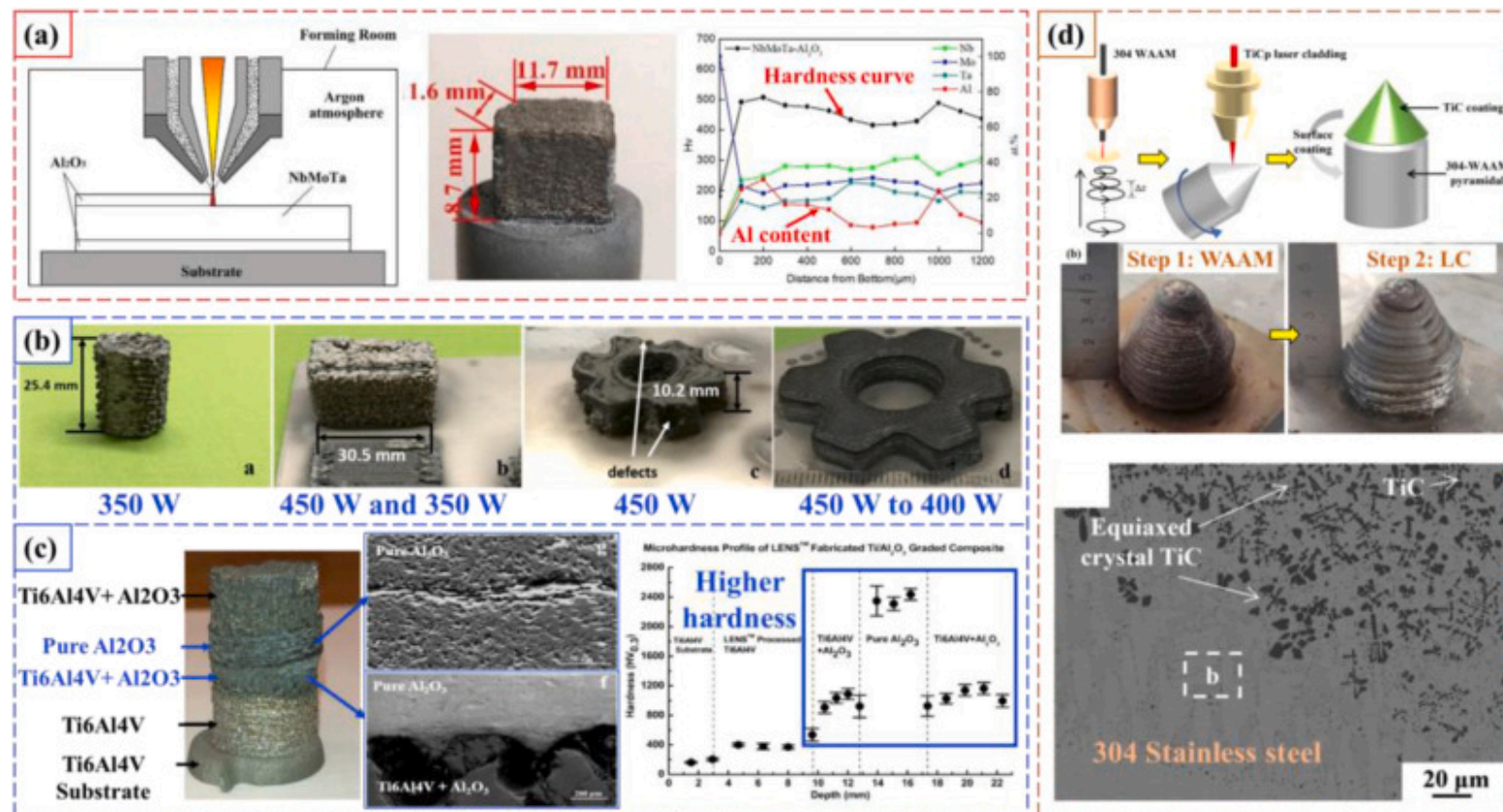
Fig. 3. CMMMS fabricated by LPBF. (a) Interfacial reaction of TiB_2 and Ti6Al4V [20]. (b) TiB_2 – Ti6Al4V laminated structure fabricated by LPBF [57]. (c1) Printed ceramic-steel interface and (c2) printed steel-ceramic-steel sandwich structure [59].

The pinning effect of the TiB nano whiskers enhanced interface bonding, leading to exceptional mechanical properties. The resulting CMMMS outperformed traditionally processed parts, with a bending strength over 2000MPa and a high bending strain of about 8%. These results highlight the potential of LPBF to produce CMMMS with enhanced mechanical properties; nevertheless, the individual phase layers are thin (<1 mm). The pronounced difference of thermal accumulation when continuously printing thick ceramic or metal parts on the counter phase can compromise printing success.

For this question, Koopmann et al. [59] demonstrated the potential of high-energy input AM to produce large ceramic or metal components on metal or ceramic substrates. They utilized LPBF to fabricate a steel-ceramic-steel sandwich structure and a laser remelting approach was employed to obtain an enhanced interface. In the process, a thin layer of ceramic powder was laid on a steel substrate and subjected to both low-power (90W) and high-power (422W) laser scans, resulting in a mechanically interlocked, serrated surface at the interface due to the Marangoni effect (Fig. 3(c1)). This interlocking was crucial for strong interface adhesion. However, challenges arose when depositing steel onto ceramic layers. The ceramic's high thermal insulation and melting points caused the steel to overheat, leading to delamination. To address this issue, the steel block added on the ceramic was designed as an open porous structure (Fig. 3(c2)) in order to reduce absolute volumetric energy input into the ceramic. This innovative design allowed them to achieve a similar mechanical interlocking interface, overcoming the challenges posed by the material's thermal properties. The optimized interface tensile strength obtained was 22 ± 4.4 MPa.

2.2.2. Multi-material DED

Direct energy deposition (DED) offers greater flexibility in switching printing material compared to PBF by delivering materials through nozzles to specified locations. This method was used by Zhang et al. [21] to fabricate Al_2O_3 – NbMoTa multilayer CMMMS with exceptional high-temperature performance (Fig. 4(a)). Every three metal layers were printed on one ceramic layer with optimized processes, resulting in CMMMS without cracks or delamination. The presence of precipitated AlTaO_4 , generated by reaction between Ta and Al_2O_3 , was believed to improve the interface bonding. Although the addition of Al_2O_3 improved hardness and wear resistance, the compressive performance of the CMMMS (608MPa) was inferior to that of pure metal parts (929MPa). This decrease was attributed to the precipitated phase; however, further experimental analysis was not conducted in this study.



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Fig. 4. CMMMS fabricated by LDED. (a) Al₂O₃-NbMoTa multilayer structure [21]. (b) Al₂O₃ parts printed on Al substrate with different laser powers, (c) CMMMS with gradient layers composed of ceramic and metal [60]. (d) Steel-TiC CMMMS fabricated by the hybrid of WAAM and laser cladding [61].

In another study, Zhang et al. [60] used DED to create a Ti- Al₂O₃ compositionally graded structure, optimizing the process particularly for Al₂O₃ ceramic materials. They addressed the challenges of low thermal conductivity and thermal shock resistance in Al₂O₃ by gradually reducing laser power layer by layer, as shown in Fig. 4(b), highlighting the importance of energy control for different layers in multi-material printing. The added Al₂O₃ also served to increase hardness. Although the ceramic layer was not directly printed onto the metal layer but rather onto a composite layer in the study (Fig. 4(c)), direct printing of CMMMS using

the method proposed is not insurmountable. Despite these efforts, the printed parts lacked precision and surface quality, which is a common challenge in DED compared to PBF.

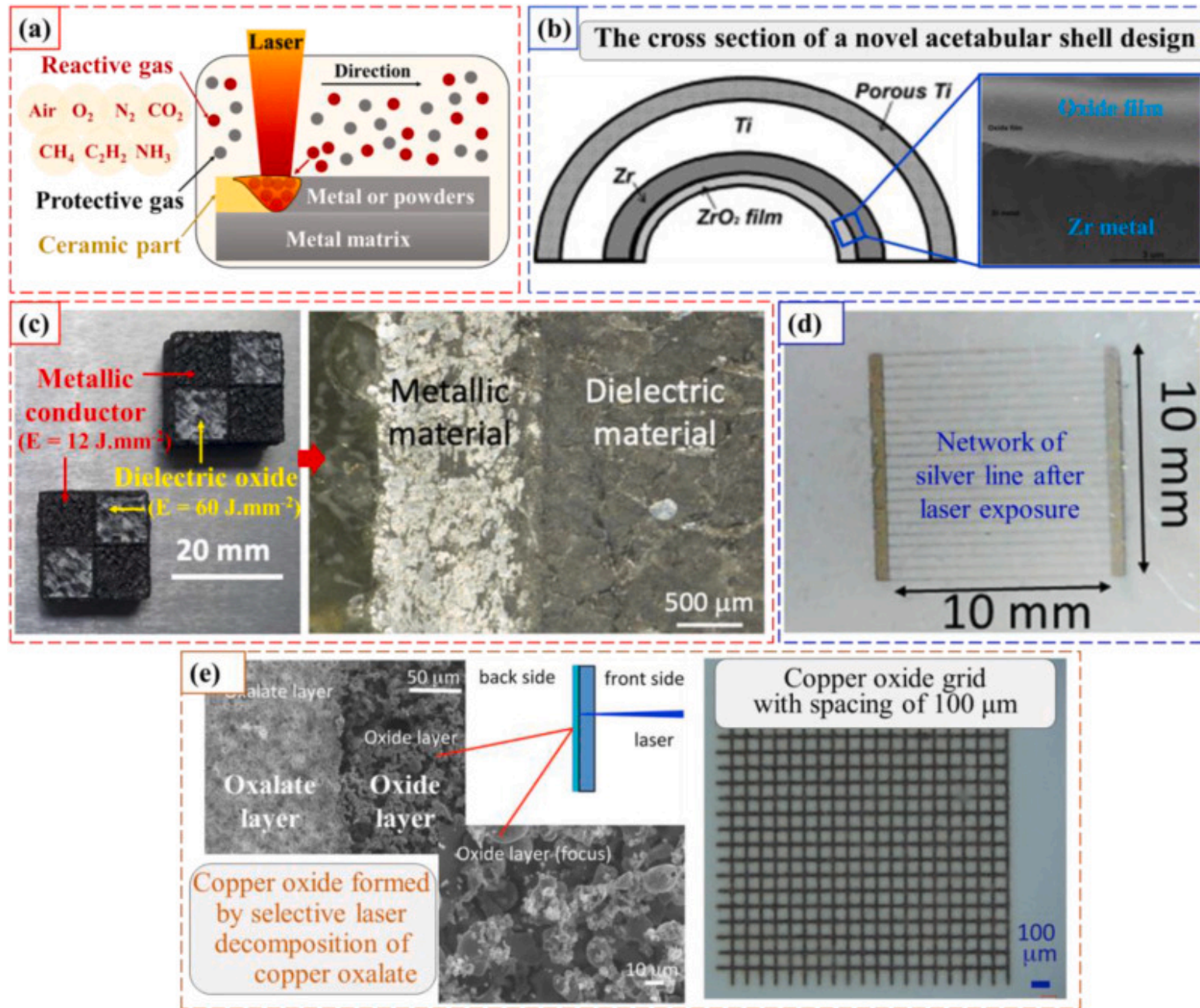
Notably, materials like aluminum are difficult to process with laser-based AM due to high reactivity and reflectivity. Wire-arc additive manufacturing (WAAM), a cost-effective DED technology based on arc welding, offers good material adaptability and multi-material printing capabilities [62]. In their process, an electric arc is used as an energy source to melt and deposit metallic wires layer-by-layer [63,64]. The physical characteristics of the arc, including medium energy density and high oxide film breaking capacity, are beneficial to the bonding of metal and ceramic materials [65]. As shown in Fig. 4(d), Niu et al. [61] demonstrated this by manufacturing a 304 stainless steel block with a TiC ceramic coating using a hybrid of WAAM and laser cladding. A strong metallurgical bond, improved hardness and wear resistance were achieved. However, surface quality remained poor, indicating the need for further research in hybrid AM technologies for CMMMS. Currently, there is limited research on WAAM of CMMMS, while many studies concentrate on the fabrication of cermet materials [66].

2.2.3. In-situ reactive multi-material AM

In-situ reactive AM is a viable method for producing CMMMS, leveraging the high-energy output of lasers to facilitate material reactions. For example, SiO powder can be sintered in air and transformed into SiO₂ by exploiting its high light absorption at 1064nm using an Nd:YAG laser [67]. This strategy is particularly beneficial for ceramics prone to microcracking during high-energy input printing process [68], potentially overcoming the inherent limitations of high-energy input AM and achieve robust bonding interfaces [20]. It offers a convenient pathway for manufacturing high-melting-point ceramics.

A common in-situ reactive printing technique involves chemical reactions between the gas atmosphere and the feed materials. To prevent metal oxidation, a protective gas atmosphere is necessary during printing. Reactive gases can then interact with the materials under laser irradiation to effect material transformation, as shown in Fig. 5(a). For instance, Balla et al. [69] created a ceramic-coated metal implant by laser oxidizing a Zr alloy (Fig. 5(b)). The choice of gas atmosphere can be customized based on specific requirements and safety considerations, including air, O₂ [69], N₂ [43,[70], [71], [72], [73], [74], [75], [76], [77]] and CH₄ [78]. Importantly, in-situ reactive printing has been applied to fabricate large multi-material structures block [70], demonstrating exceptional ultimate tensile strength and elongation rates (~1 GPa and 27% for Ti–TiN spatially heterostructure) that surpass the traditional trade-off between strength and plasticity [76]. This in-situ reactive multi-material AM method can be easily applied to LPBF and DED systems [79]. Owing to the similarity of material composition, this in-situ reactive multi-material AM method is

expected to yield robust interfacial bonding. Nevertheless, further research is needed to ensure sufficient conversion of metal materials into ceramic phases to meet the requirements of certain applications.



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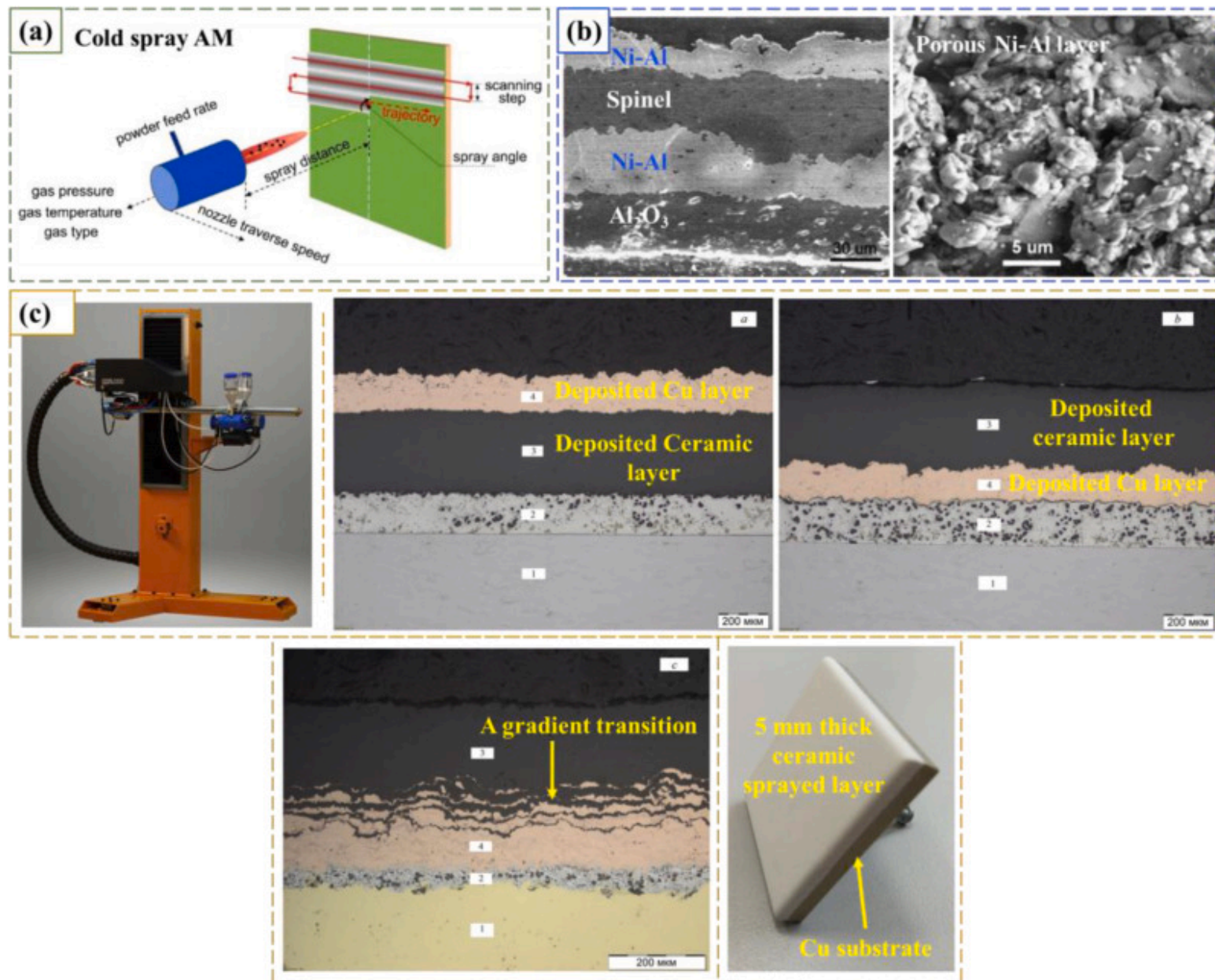
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Fig. 5. In-situ reactive multi-material AM of CMMMS based on reactive atmospheres and precursor. (a) Schematic diagram of printing process. (b) metal implant with ceramic surface [69]. (c) Printed CMMMS derived from AlSi12 alloy powders [80]. (d) Metal and (e) ceramic patterns derived from oxalates [86,89].

Another in-situ reactive printing method involves using precursor materials. Veron et al. [80] developed a technique to produce CMMMS from a single type of powder (AlSi12 alloy) by adjusting laser parameters, switching between ceramic and metal production through the dual nature of aluminum (passivation and oxidation), as shown in Fig. 5(c). However, this method has strict requirements for the properties of the printed materials. In terms of precursor material selection, it is worth noting that previous studies [[81], [82], [83], [84], [85]] have shown that metal salt precursors can be converted into metals or ceramics under specific conditions, with oxalates being a universal precursor due to their ability to form metals or oxides under laser irradiation [[86], [87], [88], [89], [90], [91], [92]], as shown in Fig. 5(d and e).

2.2.4. Multi-material SAM

Spray Additive Manufacturing (SAM) is a preferred method for creating thick coatings, as depicted in Fig. 6(a). This technology is typically categorized into thermal spray, which applies molten or semi-molten particles, and cold spray, which accelerates powder particles to supersonic speeds for coating deposition. Cold spray is especially beneficial for active metals, as it requires less heating and can be done without a protective atmosphere [93]. The sprayed conformal coatings can provide resistance to extreme erosion, heat, or electromagnetic interference [94]. The ability for high-speed deposition on non-planar surfaces makes SAM an attractive choice [95], especially for electronic devices. A humidity sensor (Fig. 6(b)) containing spinel oxide layers and Ni–Al electrode layers prepared by plasma SAM demonstrated the capability to measure humidity [96]. Optimized SAM processes, like detonation SAM, can significantly enhance interfacial bond strength, reaching up to 70MPa, and enable the fabrication of bulk CMMMS with gradient transitions [97], as shown in Fig. 6(c). In addition, it is helpful to combine SAM with other processes. For example, coatings sprayed on laser pre-oxidized surfaces are expected to have good adhesion [98].



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Fig. 6. CMMMS fabricated via SAM. (a) Schematic diagram of cold spray additive manufacturing process [22]. (b) A sprayed humidity sensor with laminated structures [96]. (c) Detonation spraying equipment and fabricated CMMMS [97].

In practice, SAM is more suitable as an auxiliary technology rather than as a stand-alone route for fabricating bulk end-use components. Owing to the limited packing density of the sprayed powders and the comparatively low energy input, SAM typically exhibits high porosity and relatively low adhesion strength. Nevertheless, the exceptional flexibility and conformal-manufacturing capability of SAM allow multiple materials to be deposited at localized regions. For example, Nai et al. [99] employed SAM to deposit a gradient composite coating that served as a filler to promote the brazing of $\text{Ti}_3\text{SiC}_2/\text{Cu}$ joints, ultimately achieving a shear strength of $\sim 110\text{MPa}$. In terms of coating fabrication, some reviews have conducted a review on SAM of metal or ceramic coatings. Refs. [100,101] reviewed studies on cold spray of metals onto ceramic surfaces and investigated the interface binding. Ref. [102] reviewed the preparation of nanostructured ceramic coatings using cold spray. Nanostructured coatings are expected to exhibit superior mechanical properties, thermal properties, electromagnetic properties, etc., compared to traditional materials.

2.2.5. Solid-state welding based multi-material AM

Traditional solid-state welding methods, including ultrasonic welding [103], diffusion welding [104] and friction stir welding [105], allow for the direct joining of metals to ceramics at temperatures below their melting points, thereby mitigating thermal stresses at the interface. Deng et al. [23] proposed an ultrasonic AM approach for fabricating multi-material structures composed of ceramic (Yttria-stabilized zirconia) and metal (Al). A $40\mu\text{m}$ thick ceramic film was laid and rolling welded onto an aluminum layer using a 9kW ultrasonic AM system, achieving a welding speed of 2.032m/min and a shear strength of 72MPa for the welded joint. Refs. [106,107] reviewed the friction-based processes for hybrid multi-material joining. Solid-state welding based AM offers valuable insights for high-energy input AM of CMMMS, while limited in creating complex 3D structures [108]. Its principal advantage lies in enabling direct ceramic-metal interfacial bonding via diffusion welding without needing fillers or transition layers. Moreover, compared to laser-based processes, the lower energy input generates gentler thermal gradients and thereby mitigates interfacial residual stresses.

2.2.6. Section summary

- (1) *Comparison of the technologies.* Based on the preceding review, different technologies exhibit variations in critical aspects such as interfacial reliability, defect tendency, and scalability. These factors determine their suitability for specific use cases.

Multi-material LPBF and DED, as the most extensively studied high-energy input AM technologies, offer complementary limitations and strengths. The rapid heating and cooling intrinsic to these two processes generate steep thermal gradients. These gradients induce residual stresses at ceramic-metal interface, compromising mechanical integrity. Moreover, the intrinsic brittleness and poor laser-matter coupling of ceramics make them prone to defects such as cracks, porosities, and rough surfaces under high-energy input [68,[109], [110], [111]]. This is partly due to the highly directional covalent or ionic chemical bonds that impede material relaxation, dislocation migration and the consolidation of ceramics [112]. Compared with DED, LPBF enables lower layer thicknesses, offering greater flexibility for tailoring interfacial bonding. As the layer-by-layer gradient adjustment of laser parameters is helpful for regulating thermal accumulation, cracking at interface is more likely to occur with thick printing layers due to thermal shock.

In addition, LPBF enables fabrication of complex geometries with better precision. In contrast, DED suffers from limited geometric accuracy, while it offers compelling advantages for large-scale manufacturing. To improve precision, integrating subtractive manufacturing process is a promising strategy. Subsequent milling, grinding or laser processing can be used to refine the rough surfaces and improve the surface quality. While such hybrid additive–subtractive strategies are already well established for metal materials [113], their application to brittle ceramics may face some challenges. The nano/micro particle beam technique proposed by Chun et al. [114] offers a potential alternative, in which high-velocity particles mechanically erode targeted regions to achieve subtractive shaping. Besides, any machining at or near multi-material interfaces should be carefully controlled, as the induced mechanical loading can substantially alter the local microstructure [115].

In-situ reactive AM presents a promising pathway to mitigate interfacial issues by promoting chemical reactions that form transition phases. The chemical reactions and products may weaken residual stresses, improve wettability, and suppress crack propagation at interfaces, thereby enhancing bonding quality. However, its scalability is currently constrained by the limited selections of precursor materials and the need for precise control over reaction kinetics. Future advancements will depend on expanding precursor

chemistries and elucidating the correlations among printing parameters, reaction mechanisms and interfacial quality. SAM is well-suited for conformal printing, but its application in bulk CMMMS fabrication is limited by high porosity and relatively low interfacial adhesion. Solid-state welding-based AM, such as ultrasonic additive manufacturing, offers the advantage of low thermal gradients and reduced residual stress, enabling robust ceramic-metal bonding. However, its capability for creating complex 3D geometries remains constrained currently.

- (2) *Interfacial bonding.* A major obstacle in quality control is improving compatibility between printed layers of different materials. Poor wettability between ceramics and metals often compromises interface bonding. In addition, mismatches in physical properties—such as melting temperature, laser absorptivity, thermal expansion coefficient, and thermal conductivity—hinder inter-material bonding [39,116]. The mismatched properties, intense physical changes result in substantial residual stresses and cracks at the interface. In-situ reaction can influence interfacial wettability and morphology through reaction products. Ductile reaction products that improve wettability, enhance load transfer, and suppress crack propagation is beneficial. In contrast, brittle phases often lead to low fracture energy interfaces. Careful control over reaction products is necessary and helpful.

Table 1 summarizes some previous research work, with the majority emphasizing process optimization as a crucial first step to improve interfacial bonding. In addition to laser parameters, the arrangement of ceramic and metal materials such as layer thickness and aspect ratio, significantly influences the strength and failure modes of interfaces [117]. Effective design of interfacial structure, roughness, and porosity can also enhance wettability at the interface [118]. However, systematic evaluations of interface quality of AM-CMMMS are still scarce. Further research is required to provide comprehensive insights into interfacial quality, residual stress distribution, and crack propagation mechanisms. Moreover, more fundamental issues, including the diffusion and migration behavior of elements during interfacial reactions, need to be elucidated to provide insights for interface regulations.

Table 1. Research on high-energy input AM of CMMMS.

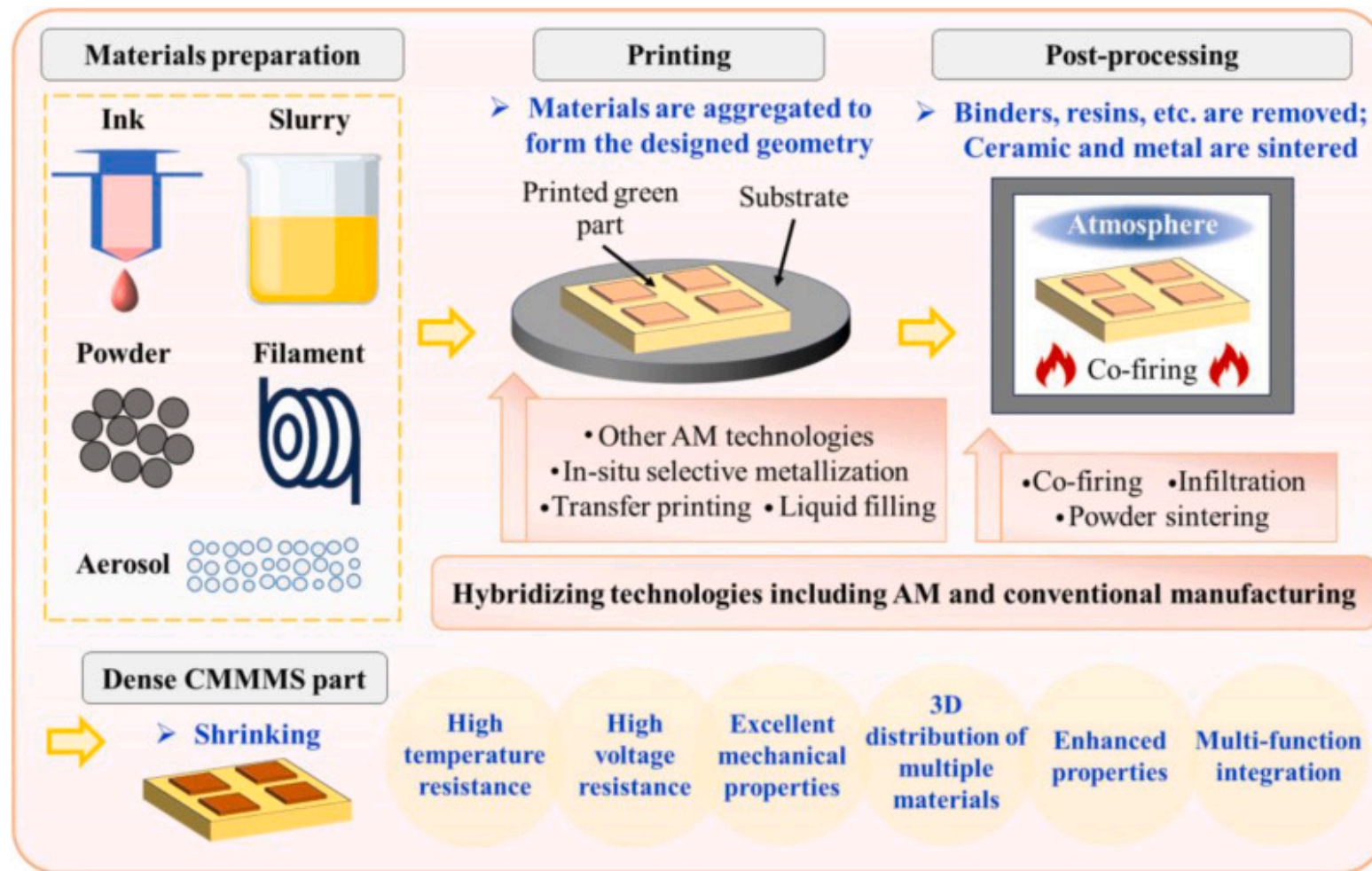
Materials	Technologies	Strategies for improving interfacial bonding	Interface strength	Properties	Ref.
ZrO ₂ /Al ₂ O ₃ , 1.2367 (X38CrMoV5-3) steel	Multi-material LPBF	Laser remelting approach at layers near interface	Tensile strength of 22±4.4MPa	Tensile strength of 20.4±4.6MPa, volumetric electrical resistance of 2.84×10 ¹⁰ Ωcm for ceramics	[59]
TiB ₂ , Ti6Al4V	Multi-material LPBF	Interface reaction	/	Ti6Al4V-to-TiB ₂ nanohardness transition (~4.8 GPa to ~16 GPa)	[20]
TiB ₂ , Ti6Al4V	Multi-material LPBF	Process optimization	/	/	[56]
TiB ₂ , Ti6Al4V	Multi-material LPBF	Process optimization and interface reaction	/	Layered TiB ₂ -Ti6Al4V structures with bending strength exceeding 2000MPa and ~8% strain	[57, 119]
ZrO ₂ /Al ₂ O ₃ , pure Al, pure Cu for battery cell cap fabrication	Multi-material LPBF	Strategy 1: overlapping in the transition zone; Strategy 2: remelting the transition zone; Strategy 3: stepped interface at the transition zone; Strategy 4: lowered volumetric energy density near the transition zone	/	Dielectric breakdown strength of 7.92 Kv/mm for ceramic; Average conductivity of 30.87MS/m for pure Al, 40.01 MS/m for pure Cu	[120]

Materials	Technologies	Strategies for improving interfacial bonding	Interface strength	Properties	Ref.
Al ₂ O ₃ , NbMoTa	Multi-material LDED	Process optimization	/	Improved hardness (462.5±43.5 HV0.5) for metal; Improved wear resistance (by 55%) for metal; Reduced compression performance: yield strength of 502MPa, ultimate strength of 608MPa, and compression rate of 3.25%; Electrical anisotropy	[21]
Al ₂ O ₃ , Ti6Al4V, Ti6Al4V+Al ₂ O ₃ composites	Multi-material LDED	Process optimization: gradually changing energy input at layers near the interface; Composition transition layers	/	Improved hardness (1000 HV0.3 of the Ti6Al4V+Al ₂ O ₃ sections)	[60]
TiC, 304 stainless steels	Hybrid of WAAM and laser cladding	Optimized process	/	Printed 304 stainless steel block with high hardness (654.7±41.0 HV _{0.2}) and high wear resistance (reduced by about 99%) TiC coating	[61]
ZrO ₂ , Al	Solid-state welding	Optimized process	Shear strength of 72MPa (as-welded) and	/	[23]

Materials	Technologies	Strategies for improving interfacial bonding	Interface strength	Properties	Ref.
	based multi-material AM		103 MPa (heat-treated)		
Ti and N ₂ for fabricating Ti–TiN composites	In-situ reactive multi-material AM	Optimized process and interface reaction	/	Excellent ultimate tensile strength (~1 GPa) and elongation rates (27%) for Ti–TiN spatially heterostructure	[76]
Ti6Al4V and N ₂ for fabricating Ti6Al4V–TiN composites	In-situ reactive multi-material AM	Optimized process and interface reaction	/	Improved microhardness (1141 HV0.2), ultimate tensile strength (1071.4 MPa) and elongation (15.6%) of Ti6Al4V–TiN composites	[79]

2.3. Low-energy input AM

Compared with high-energy input AM, low-energy input AM tends to maintain the geometric accuracy of the printed green body [121]. Additionally, lower temperature gradients during heat treatment (debinding and sintering) reduce the occurrence of defects. These technical features make AM highly suitable for the preparation of electronic devices. The flexibility of AM enables the creation of more miniaturized and refined CMMMS, potentially revolutionizing the design and fabrication of electronic devices by overcoming the limitations of traditional manufacturing methods. Notably, the fabrication process often combines AM with conventional technologies (as shown in Fig. 7), a concept known as hybrid AM or hybrid 3D printing. Low-energy input AM technologies discussed in Section 2.3 may integrate various AM processes but do not include the integration of traditional manufacturing techniques; they are categorized based on the printing feedstocks. The hybridization of AM and conventional manufacturing will be discussed in Section 2.4.



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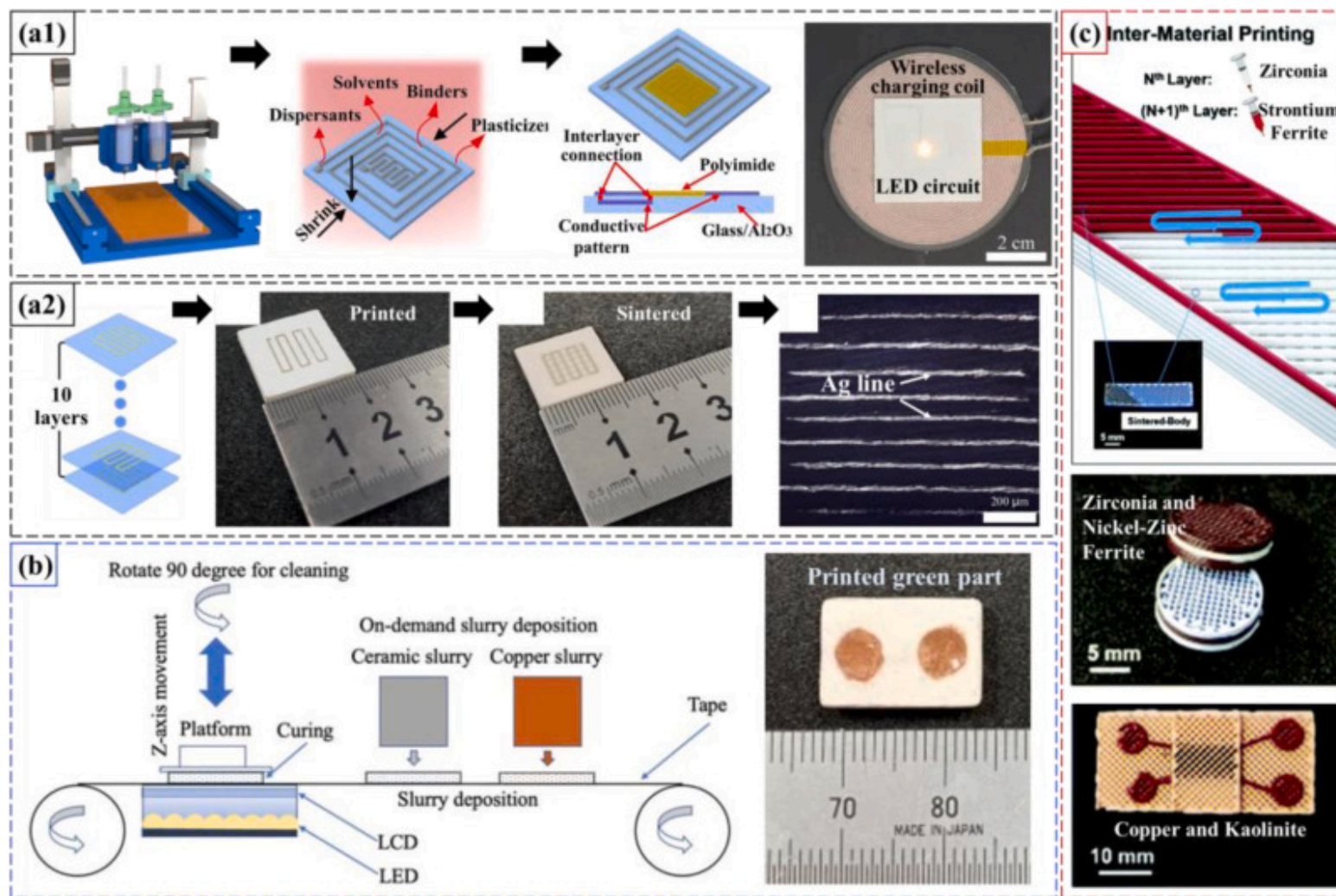
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Fig. 7. An overview of low-energy input AM and hybrid AM for preparing CMMMS.

2.3.1. Based on ink or slurry

Low-energy input AM of CMMMS based on liquid materials has received extensive research attention, with common techniques including MJ, VP and their hybrids, employing ink or slurry as printing feedstocks. An important issue in these processes is

preventing contamination between different materials, especially liquids. While a rotating platform solution has been proposed [122], employing multiple nozzles is still a more efficient and common solution for this issue. Duan et al. [2] utilized a direct ink writing (DIW) 3D printer with dual extrusion heads (Fig. 8(a1)) to fabricate ceramic ($\text{Al}_2\text{O}_3/\text{glass}$) and metal (Ag) Low-Temperature Co-fired Ceramic (LTCC) circuits. The distinct inks were deposited precisely, resulting in a $500\mu\text{m}$ wide Ag line and a well-bonded 10-layer LTCC after co-firing at 850°C for 15 min, as shown in Fig. 8(a2). Wang et al. [123] employed a similar approach to create 3D LED circuit boards and ceramic heaters/thermometers, which accurately measured the temperature of a 10wt.% H_2SO_4 solution, showcasing the advantages of AM in component layout. In ink-based MJ printing technologies, strict limitations are imposed on the rheological properties such as viscosity to prevent ink intermixing and collapse of the printed structure. The viscosity range of liquid slurries allowed by VP is relatively wide because the material at specified locations will be photopolymerized to form a solid green body.



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Fig. 8. CMMMS fabricated by low-energy input AM based on liquid materials. (a1) Multi-material DIW technique for fabrication of CMMMS and (a2) printed LTCC [2]. (b) Multi-material VP [124]. (c) Universal self-curable techniques for CMMMS [125].

Jiang et al. [124] developed a novel 3D printing technology based on bottom-up VP, where ceramic and metal (Cu) slurries are not directly deposited in the printing area, but rather selectively deposited onto a transparent film, as shown in Fig. 8(b). The

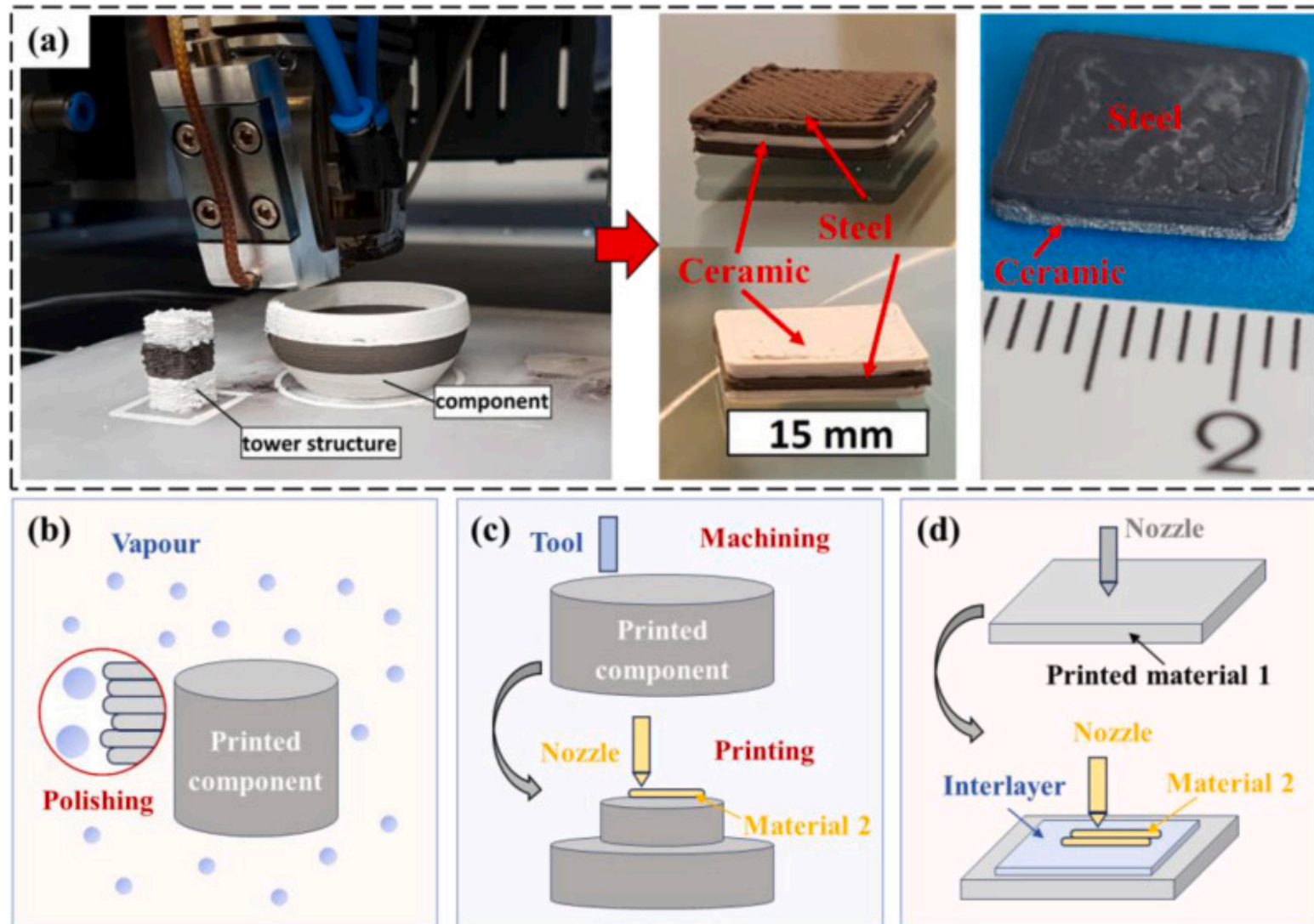
material on the film was photocured as the film moved over the substrate or the already cured part. The shrinkage rates of ceramic and metal pastes were comparable at 67.97% and 68.85%, respectively, which ensured that the bulk metallic structure (embedded metallic tunnel with a diameter of 5 mm) would not separate from the ceramic part due to significant shrinkage mismatch. Raynaud et al. [24] utilized Stereolithography (SL) to print ceramic parts and deposit conductive metal (Ag) patterns via DIW, removing uncured ceramic slurry through a cleaning apparatus before metal deposition. The same process was also employed in HTCC fabrication [3,126], yielding an alumina ceramic substrate with a Young's modulus of 356 ± 23 GPa [3].

VP offers high precision and efficiency but faces challenges with materials of high refractive index or absorption rate, including most metals, nitrides, carbides and black oxides [127]. Developing technologies with lower requirements for slurry or ink properties is crucial for expanding the capability and manufacturing range of multi-material AM. Zhang et al. [125] overcame this by developing an improved epoxy-amine self-curing system for CMMMS, allowing self-curing under ambient conditions with the potential for accelerated curing through heating (Fig. 8(c)). This technique successfully printed a copper-kaolin multi-material structure for electronic sensors, highlighting its universal applicability in printing various metal and ceramic materials. The simultaneous improvement of material diversity and structural complexity is of great significance for advanced multi-material devices.

The studies above elucidate the capability of low-energy input AM based on liquid materials in fabricating CMMMS electronic devices. In Section 3.2, The details of the breakthroughs achieved by these technologies in the manufacturing of CMMMS for electronic devices will be provided.

2.3.2. Based on filament or powder

In the Fused Filament Fabrication (FFF) process, thermoplastic filaments are heated and extruded through a nozzle, then cooled and solidified at their deposition points. This method's independence from powder optical properties makes it versatile for various ceramic and metal materials. Johannes et al. [9] explored the use of FFF for zirconia-stainless steel multi-material structures (Fig. 9), addressing material preparation, equipment development, and printing processes. While their study validated the potential of FFF for producing CMMMS, it did not include property testing, and the surface quality of the parts was poor.



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Fig. 9. (a) CMMMS fabricated by low-energy input AM based on filament materials [9]. (b) Vapour polishing, (c) subtractive manufacturing and (d) inserting interlayers methods for improving printing accuracy.

FFF offers compelling advantages in cost and scalability, yet inferior surface accuracy impedes its adoption. A rough surface will result in discontinuous metal wires deposited on it. Breakthroughs in improving printing accuracy would constitute a major advance. Limited accuracy stems primarily from nozzle diameter constraints and the stair-step effect inherent in layer-by-layer deposition. Beyond parameter optimization, some ideas have been proposed to improve the forming accuracy. Vapour polishing process has been widely used [128], in which vapour is used to soften the outer layers of the printed part to achieve a polishing effect, as shown in Fig. 9(b). However, how to integrate this process with multi-material AM processes is a problem. The second approach is to repair the printed structure by combining subtractive manufacturing (Fig. 9(c)), where in-situ monitoring can assist in precisely identifying the areas to be removed [129]. Defect identification and process planning are two important issues for this approach. As shown in Fig. 9(d), the third approach is to improve the smoothness by printing interlayer materials on the surface of FFF-printed parts [130], where the coaxial printing method is promising [131].

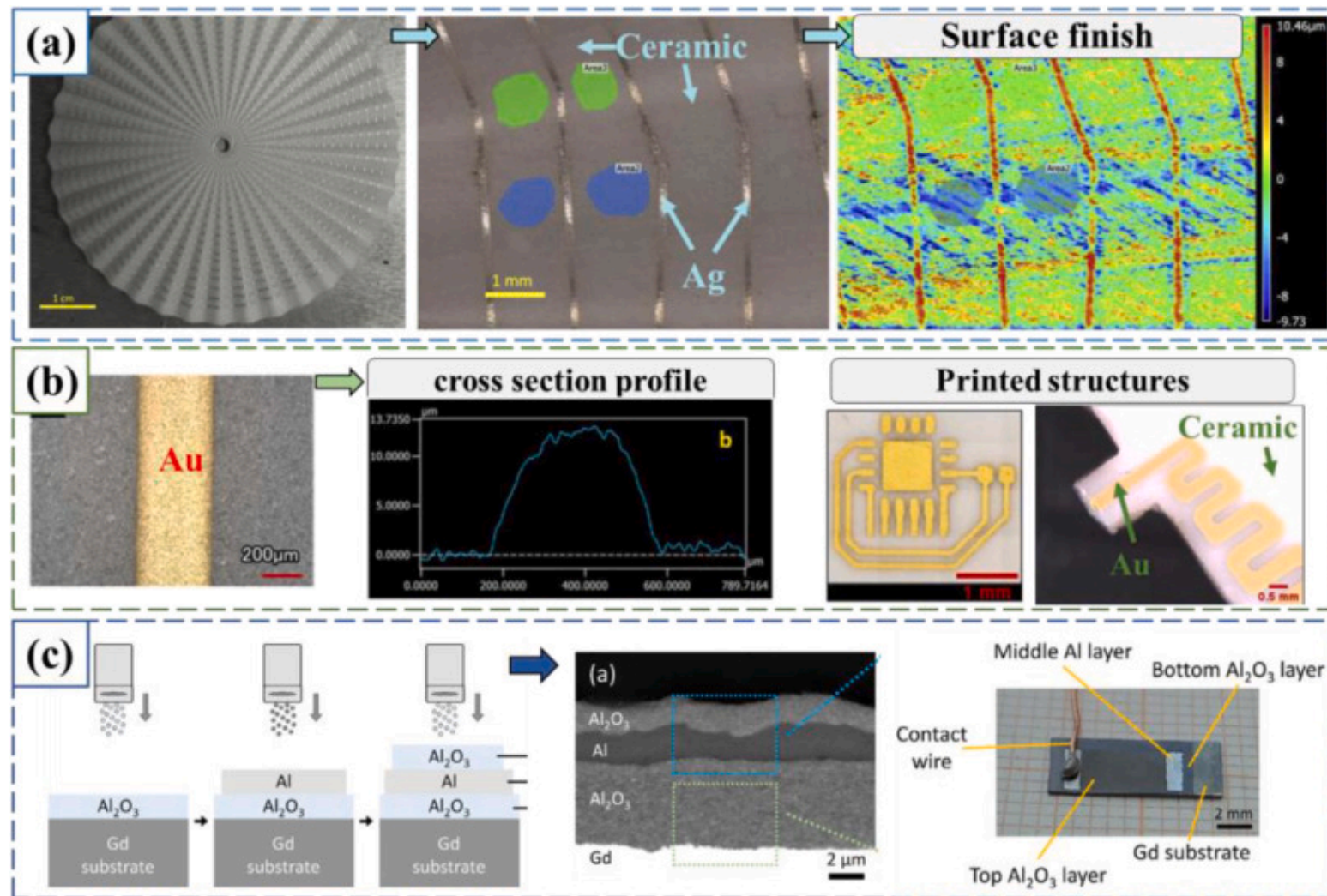
Binder Jetting (BJ), which uses powders and binder jetting for part formation, has seen significant development [132]. It has been effectively used in the AM of metals [133] and ceramics [134,135]. However, its application in CMMMS is still nascent. Schulz et al. [136] combined BJ for ceramic structures with inkjet printing (IJP) for metal patterns to create CMMMS. This approach also enabled the surface mounting of electronic devices for the integration of passive components and semiconductors. Further research is needed to explore the potential of BJ in CMMMS fabrication. Owing to its reliance on binders to agglomerate dispersed powders, BJ has a wide range of applicability to different materials; however, this same printing mechanism results in limited density, with values exceeding 90% already considered high [137]. Further improvement demands a deeper, system-level understanding of interactions among the binder, powder bed, and process parameters throughout the build.

2.3.3. Based on aerosol

Aerosol Jet Printing (AJP) involves dispersing solid or liquid particles in a gas medium and is widely applied in electronic devices [13,138]. This technique atomizes materials and mixes them with a carrier gas before jetting them onto complex surfaces, such as the conformal printing of metal wires on all faces of a ceramic cube [139].

Oh et al. [140] utilized IJP to preliminarily print large-sized zirconia dielectric substrates, followed by AJP of silver to create double helix arms with sinusoidal wave-like pattern for microstrip antennas applications. A surface roughness of less than 5 μm was achieved, as shown in Fig. 10(a). Alhendi et al. [25] employed digital light processing (DLP) to fabricate alumina dielectric substrates, and subsequently create metal patches via AJP, ultimately producing broadband microstrip antennas (Fig. 10(b)). AJP

has also enabled the fabrication of high-performance CMMMS microwave and millimeter-wave devices through hybrid AM techniques [141,142].



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Fig. 10. CMMMS fabricated by low-energy input AM based on aerosol. (a) Sinusoidal spiral antenna printed by IJP and further printed with an aerosol jetted silver spiral [140]. (b) Printed Au line and ceramic-Au structures [25]. (c) Multilayer Al₂O₃-Al

structure [143].

In addition to printing surface structures, AJP can also be used for the preparation of multilayer CMMMS, such as $\text{Al}_2\text{O}_3/\text{Al}/\text{Al}_2\text{O}_3/\text{Gd}$ multilayers, with defect-free layers and good connectivity [143], as shown in Fig. 10(c). In general, aerosol jetting printed wires can maintain good performance. The width of printed metal lines was less than $20\mu\text{m}$ [139,140]. The printed silver conductors exhibited a conductivity of $2.1 \times 10^{-7} \text{ S/m}$ at 49.2GHz [142]. In fact, AJP tends to be an excellent choice for non-planar conformal printing of thick films, rather than for manufacturing bulk structures. Similar to SAM, AJP can be considered as a potent complementary technique in hybrid AM.

2.3.4. Section summary

Low-energy input AM is predominantly utilized in the electronics sector for CMMMS. Critical performance metrics for electronic devices include routing density, conductivity, dielectric strength, constant, and loss [46]. Aerosol Jet Printing (AJP) can achieve metal line widths of $10\text{-}20\mu\text{m}$ [49,139,140], finer than the $30\mu\text{m}$ lines produced by MJ [144]. Despite the theoretical possibility, VP is rarely used for fine metal conductors in CMMMS due to material removal challenges, and is commonly used for ceramic substrate fabrication. In fact, AM is competitive with traditional manufacturing, with devices like antennas and microwave components often outperforming or matching those made by conventional methods [145]. For example, MJ-printed microstrip patch antennas with the design frequency at 9–11 GHz demonstrated a high gain of 7.52dB [146]. DIW-printed platinum films showed lower porosity and superior precision, stability, and monitoring capabilities compared to sputtered films [147]. Post-processing such as chemical polishing can further improve the quality of the printed structures [148]. Overall, the properties of printed electrical structures are able to generally meet usage requirements. Table 2 provides an overview of devices printed via low-energy input AM and their properties.

Table 2. Examples of low-energy input AM of CMMMS and their properties.

Feedstocks	Technologies	Applications	Metal line width	Properties	Ref.
$\text{Al}_2\text{O}_3/\text{glass}$ ink, Ag ink	DIW	LTCC, multilayer LED circuit	$500\mu\text{m}$	Dielectric constant: 7.65; Dielectric loss tangent: 1.54×10^{-3} at 3GHz; Flexural strength of the ceramic: 186.2MPa	[2]

Feedstocks	Technologies	Applications	Metal line width	Properties	Ref.
LTCC aqueous slurry, commercial Ag slurry	DIW	3D LED circuit boards, ceramic heaters/thermometers	/	Resistivity: $2.34 \pm 0.04 \mu\Omega\text{cm}$; Temperature coefficient of resistance: $3.56 \times 10^{-3} \text{ }^\circ\text{C}^{-1}$	[12 3]
LTCC slurry, Cu slurry	VP	LTCC	/	Average hardness of the ceramic: 537.1 HV; Flexural strength of the ceramic: 126.61 MPa; Electrical resistance: 0.175Ω	[12 4]
Al_2O_3 /glass slurry, Ag slurry	Hybrid SL and DIW	LTCC	$110 \mu\text{m}$	Young's modulus of the ceramic: $83 \pm 24 \text{ GPa}$; Flexural strength of the ceramic: $84 \pm 24 \text{ MPa}$; Permittivity: 5.05; Dielectric loss tangent: 1.8×10^{-3} ; Resistivity: $\sim 2.8 \times 10^{-8} \Omega\text{m}$	[24]
Al_2O_3 slurry, tungsten slurry	Hybrid SL and DIW	HTCC	$110 \mu\text{m}$	Young's modulus of the ceramic: $356 \pm 23 \text{ GPa}$; Resistivity: $\sim 6.9 \times 10^{-7} \Omega\text{m}$	[3]
Cu slurry, kaolinite slurry, ZrO_2 slurry, etc.	DIW with an improved epoxy-amine self-curing method	Electronic sensor and other CMMMS	/	Resistivity: about $5.0 \times 10^{-8} \Omega\text{m}$; Flexural strength of ZrO_2 : $\sim 340 \text{ MPa}$	[12 5]
ZrO_2 and stainless steels filament	FFF	Heating devices	/	/	[9]
Al_2O_3 /glass mixed powder, Ag ink	Hybrid BJ and IJP	LTCC	1 mm	Flexural strength of the ceramic: 242 MPa; Dielectric constant: 8.5; Dielectric loss tangent: 2×10^{-3} at 1 GHz; Electrical X–Y conductivity: 31490 S/m; Vertical conductivity: 1605 S/m	[13 6]

Feedstocks	Technologies	Applications	Metal line width	Properties	Ref.
ZrO ₂ ink, Ag aerosol	Hybrid AJP and MJ	Sinusoidal spiral antenna	~150μm	Bulk relative permittivity: 23; Loss tangent: 1.3×10^{-3} at microwave frequencies	[140]
Al ₂ O ₃ slurry, Au aerosol	Hybrid AJP and DLP	Conformal broadband antenna	~50μm	The antenna was around 8 dBi for most frequencies within 2 to 18GHz	[25]
Al ₂ O ₃ slurry, Ag aerosol	Hybrid AJP and SL	Millimeter wave devices with singular shapes	16.5μm	The achieved transmission lines showed a return loss better than 10dB, an insertion loss better than 0.8dB up to 60GHz. Conductivity of the printed Ag layers: 2.1×10^7 S/m at 49.2GHz.	[142]

Notably, co-sintering technologies of CMMMS warrant research. On the one hand, co-sintering of large-volume CMMMS introduces challenges in shrinkage matching and interfacial evolution. On the other hand, the development of novel sintering technologies remains to be explored. Microwave sintering leverages the interaction between materials and the microwave electromagnetic field, converting electromagnetic energy into thermal energy [149]. This method heats the component uniformly, with reduced sintering temperatures and temperature gradients [4], possibly accommodating different sintering temperature requirements for metals and ceramics. Cold sintering, introduced by Randall's team, is another novel approach to achieve densification at low temperatures (below 350°C) [150,151]. Galotta et al. [152] detailed the characteristics and mechanism of cold sintering, noting its success for piezoelectric and dielectric ceramics for electronic devices. Additionally, flash sintering, based on irradiation heating, has enabled ultra-fast sintering of ceramics, cermet, and metals [[153], [154], [155]]. These innovative sintering methods may provide new insights into improving interfacial bonding between ceramics and metals.

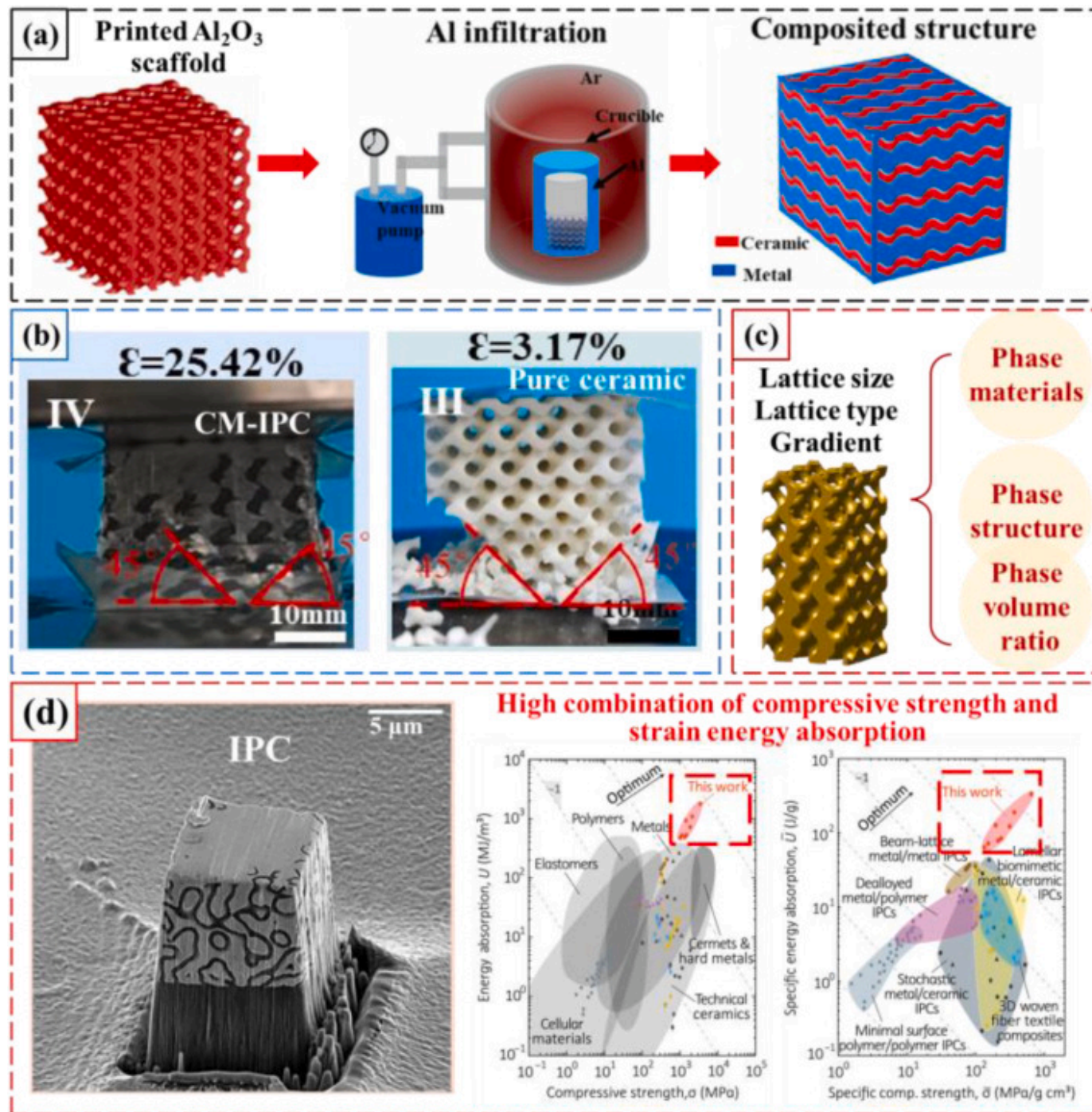
2.4. Hybrid of AM and conventional manufacturing

The complete process of low-energy input AM of CMMMS encompasses both the printing and post-processing stages, as shown in Fig. 7. During these stages, traditional manufacturing techniques such as chemical plating, vapour deposition, direct copper bonding, thermal evaporation, and sputtering can be integrated to achieve complex structures and enhance the control over

structural geometry and material distribution. This hybrid approach compensates for the limitations of AM, particularly the miniaturization required for electronic components and the shrinkage issues during heat treatment. Imanaka et al. [156] demonstrated the feasibility of fabricating CMMMS by hybrid aerosol deposition, electroplating Cu plating, sputtering Cu sputtering, and conductor etching. Such hybrid methods not only improve the functionality of CMMMS but also pave the way for the production of more complex, multifunctional devices. Several representative technologies are introduced below.

2.4.1. Hybrid of AM and infiltration

Ceramic-metal interpenetrating phase composite (CM-IPC) is composed of continuous ceramic and metal phases, which are topologically interconnected throughout the structure, as shown in Fig. 11(a). The CM-IPC fabrication process utilizing AM and infiltration generally consists of two steps: creating a porous scaffold and infiltrating it with secondary phase materials. Ceramic materials and molten metal liquids are commonly chosen as the scaffold material and infiltrating counter phase respectively, attributing to the inherent differences in melting points. Compared to traditional manufacturing techniques like freeze casting [157], CM-IPC fabricated by AM allows for the orientation of scaffold structures to achieve desired performance characteristics. The following sections will introduce CM-IPC based on their targeted properties.



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Fig. 11. (a) Basic process flow of preparation of CM-IPC [158]. (b) Improving the properties of CM-IPC through structural design [158]. (c) Structural design factors influencing the performance of CM-IPC. (d) Nano-architected CM-IPC [26].

CM-IPC with enhanced mechanical properties: Research on CM-IPC primarily concentrates on their mechanical properties, with structural ceramics like Al_2O_3 and ZrO_2 being commonly used. The metallic phase in CM-IPC enhances ductility, while the ceramic phase ensures strength. During compression, CM-IPC exhibits densification as the fractured ceramic is encapsulated by the metal, becoming denser [158]. AM has shown potential in creating CM-IPC with exceptional performance. Singh et al. [159] reported a ZrO_2 -Al interpenetrating phase composite (IPC) with 400 times higher specific energy absorption than pure ceramic scaffolds. Given the paramount importance of overall mechanical performance, a broader tolerance for surface quality (e.g. surface roughness) is acceptable, enabling the use of less precise AM techniques like WAAM. Moreover, structural design plays a crucial role in enhancing CM-IPC performance, with mechanical interlocking reinforcing bond strength between heterogeneous materials and mitigating concerns over interface bonding defects.

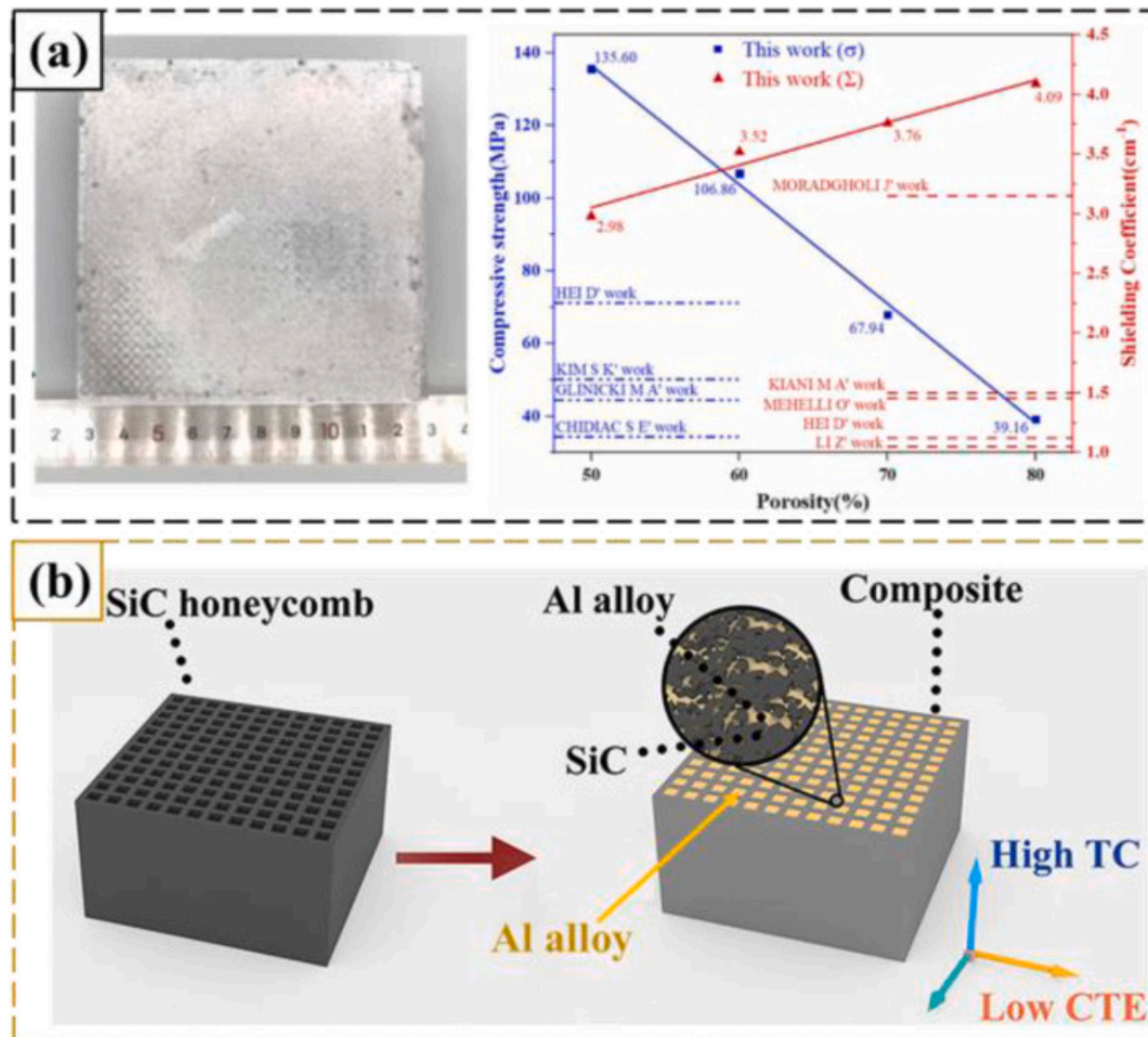
Lu et al. [158] discovered that the ultimate crack direction in CM-IPC under compression aligns with the original alumina scaffold (Fig. 11(b)), emphasizing the significant influence of the ceramic scaffold structure on CM-IPC performance. Scaffold structures initially impact the properties of the scaffold and subsequently influence the CM-IPC performance [160,161]. CM-IPC with Triply Periodic Minimal Surface (TPMS) structures have shown increased stiffness, strength, and toughness over other architectures, mainly due to their uniform stress distribution [162,163]. Consequently, extensive studies have focused on TPMS-based CM-IPC.

Fig. 11(c) outlines the structural design factors affecting CM-IPC performance. For example, increasing pore size improved the quasi-static compression and resistance performance of the Al_2O_3 -Al IPC, contrasting with the behavior of Al_2O_3 scaffolds [164]. Chandrasekaran et al. [165] controlled the hardness of B_4C -Al IPC by adjusting lattice spacing and ceramic phase content. Gradient design capability of AM allows for the creation of structures with varying mechanical properties. Li et al. [166] demonstrated that Al_2O_3 -Al IPC with larger gradients exhibited better compressive strength and energy absorption compared to uniform ceramic scaffolds. It is evident that the performance changes of CM-IPC and pure ceramic scaffolds showed opposite trends, which are related to the proportion of different phase contents. Therefore, it is necessary to consider the effect of lattice size on the proportion of the phase content. Singh et al. [167] and Del Frari et al. [168] found that phase structure and volume

fraction significantly influence IPC performance, especially under static loads and when the mechanical properties of the phases differ significantly. The availability, orientation and shape of a low-compliance path for stress transfer through the stiffer phase were believed to be crucial for macroscopic performance. These insights provide a basis for designing CM-IPC by considering phase materials, structures, and volume ratios.

Notably, CMMMS with microscale or nanoscale is expected to exhibit remarkable performance and tunable properties, such as mechanical and thermal behavior [169]. Bauer et al. [26] developed a method for manufacturing nanostructured CM-IPC (Fig. 11(d)). The prepared IPC showed extremely high compressive strength and strain energy absorption, with stresses reaching 3.5 GPa, surpassing ceramic-based composites while maintaining large strain deformation capabilities. The study validates the special properties of nanostructures due to the size effect [170,171]. Furthermore, the self-assembly of spinodal architectures may offer a path to produce macroscopic structures with preserved nanoscale benefits [172].

Functional CM-IPC: CM-IPC with special functions can be achieved by selecting appropriate functional ceramic materials for the scaffold. For instance, a neutron shielding IPC with variable shielding coefficients was created by infiltrating B₄C ceramic-epoxy resin into LPBF-printed AlSi10Mg TPMS structures (Fig. 12(a)) [173]. The piezoelectric CM-IPC demonstrated a d_{33} constant approximately 7 times greater than traditional materials, due to effective load transfer and improved polarization from the aligned structure [174]. Such properties allow for sensing, with the metallic phase providing resistance to mechanical impacts and acting as an actuator, offering additional functionality in a small space [175]. These multifunctional IPCs are crucial for military, space, and other high-demand applications, though further research is necessary.



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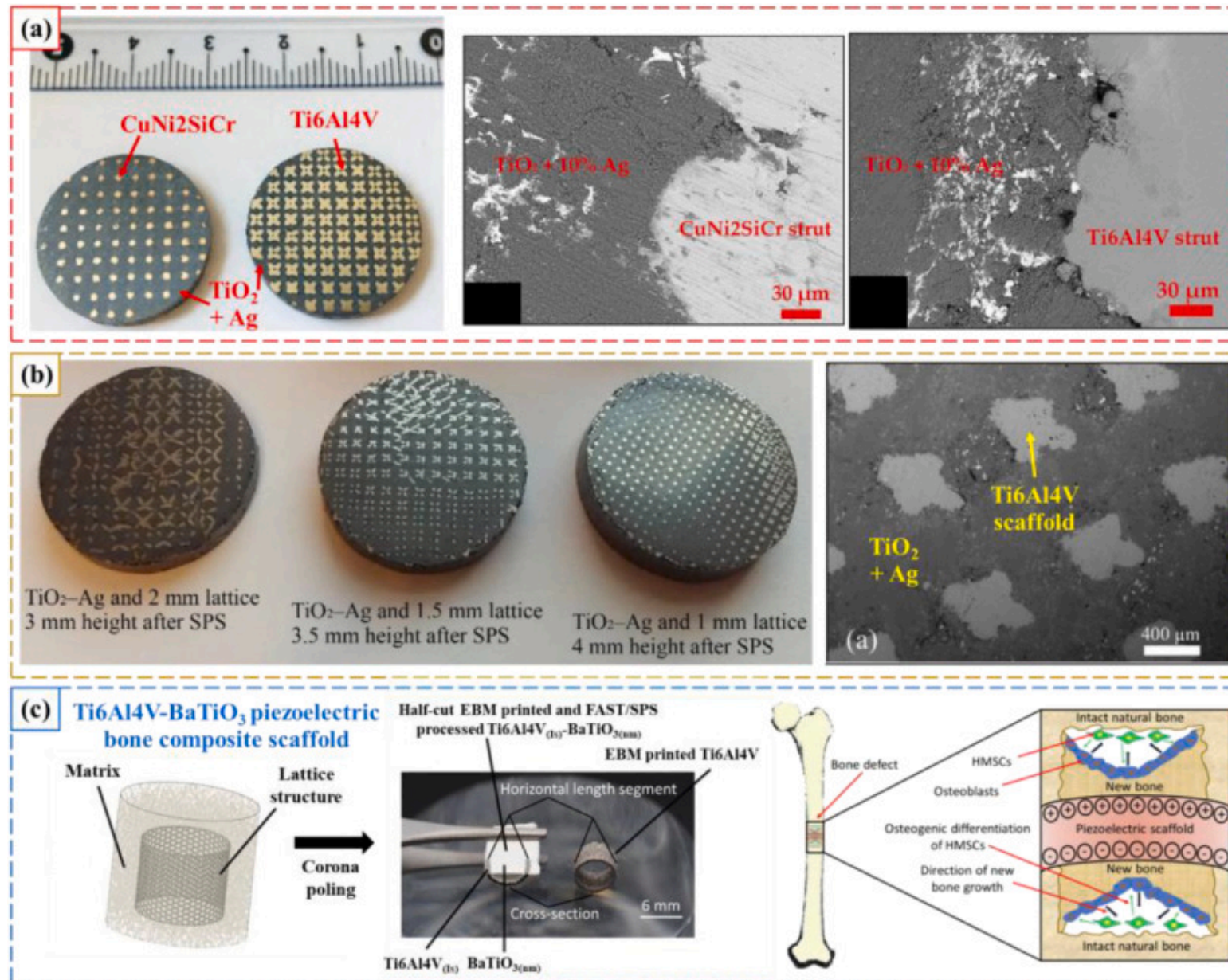
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Fig. 12. Design and fabrication of IPC with special functions. (a) IPC with neutron shielding function [173]. (b) SiC–Al IPC with anisotropic heat dissipation performance [27].

In thermal management, CM-IPC is favored for applications like automotive brake rotors due to its combination of mechanical and thermal properties [176]. SiC ceramics, known for higher thermal conductivity than alumina, are often used in thermal management devices [27]. AM allows for precise control over structure to achieve lower thermal expansion and anisotropic heat dissipation [177,178], as shown in a SiC honeycomb with directional pores (Fig. 12(b)). This customization is essential for high-performance thermal management in miniaturized electronics. Nanostructured CM-IPC for functional applications also holds promising potentials in thermal management and electromagnetic modulation [179], as their high surface area and more tortuous charge transport pathways may have a great impact. However, further research is required to fully exploit these potentials. Future integration of diverse functional materials such as shape memory materials is also poised to unlock unprecedented performance breakthroughs.

2.4.2. Hybrid of AM and powder sintering

Powder sintering is an alternative to melt infiltration in the fabrication of CM-IPC. This method involves filling 3D printed metal scaffolds with ceramic powder, which is then sintered using conventional powder sintering techniques [180]. Such CM-IPCs have potential applications in biomedicine, electro-thermal applications, and refractory materials [181,182]. The ceramic filling can enhance both the mechanical properties and biocompatibility of metal implants [183]. As shown in Fig. 13(a and b), CM-IPCs with antibacterial properties can be achieved by incorporating Ag or Cu nanoparticles into the ceramic material [28,184]. Piezoelectric ceramics in CM-IPCs can also promote cell-material interactions and tissue regeneration under electrical stimulation [185], as shown in Fig. 13(c). Rahmani et al. [183] investigated the possibility of introducing cracks or pores in the ceramic phase for drug delivery while maintaining strength. In the future, it is possible to study IPCs composed of various ceramic or metal materials to leverage the ability of AM to integrate complex functions.



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Fig. 13. CM-IPC composed of printed metal lattice and filled sintered ceramic powder: (a-b) CM-IPC with antibacterial and antiviral properties [28,184]. (c) Piezoelectric CM-IPC for promoting tissue regeneration [185].

However, this approach has its challenges. The sintering temperature must be below the metal scaffold's melting point, and high temperatures may alter the metal scaffold's internal phase composition. Additionally, ceramic powder shrinkage during sintering could negatively impact the multi-material structure's consistency and performance.

Unlike particulate composites that primarily increase hardness and wear resistance with limited toughness improvement [186], the distinct 3D continuous network structures for each phase in CM-IPC preserve their individual properties and enable efficient transmission of internal loads, such as mechanical forces and electrons, leading to multifunctional properties and synergistic enhancements. Table 3 summarizes some examples of CM-IPC prepared via AM and their properties.

Table 3. Examples of CM-IPC and their properties.

Ceramic-metal	Properties	Ref.
Al ₂ O ₃ -Al	Quasi-static compressive strength: 89.06MPa, approximately 6 times that of the ceramic scaffold; Energy absorption capacity: 2569.16kJ/m ³ , 15 times that of the ceramic scaffold.	[158]
Al ₂ O ₃ -Al	Compressive strength: 176.1 MPa, 2528.4% higher than that of the pure ceramic scaffold; Energy absorption: 4.638 MJ/m ³ , 13598.2% higher than that of the scaffold	[166]
SiO ₂ -Al	Compressive strength: 165.3MPa; Specific compressive strength: 75.23kNm.kg ⁻¹ , 169% of monolithic metal phase; Fracture toughness: 12.89MPa.m ^{1/2}	[161]
ZrO ₂ -Al	Compressive strength: 220MPa; Specific energy absorption: 28.9J/g, 400 times higher than that of pure ceramic scaffolds	[159]
SiC-Al	Anisotropic heat properties: axial direction: thermal conductivity 138.3W/(m·K), coefficient of thermal expansion 12.01 × 10 ⁻⁶ /K; radial direction: thermal conductivity 100.5W/(m·K), coefficient of thermal expansion 11.40 × 10 ⁻⁶ /K	[27]

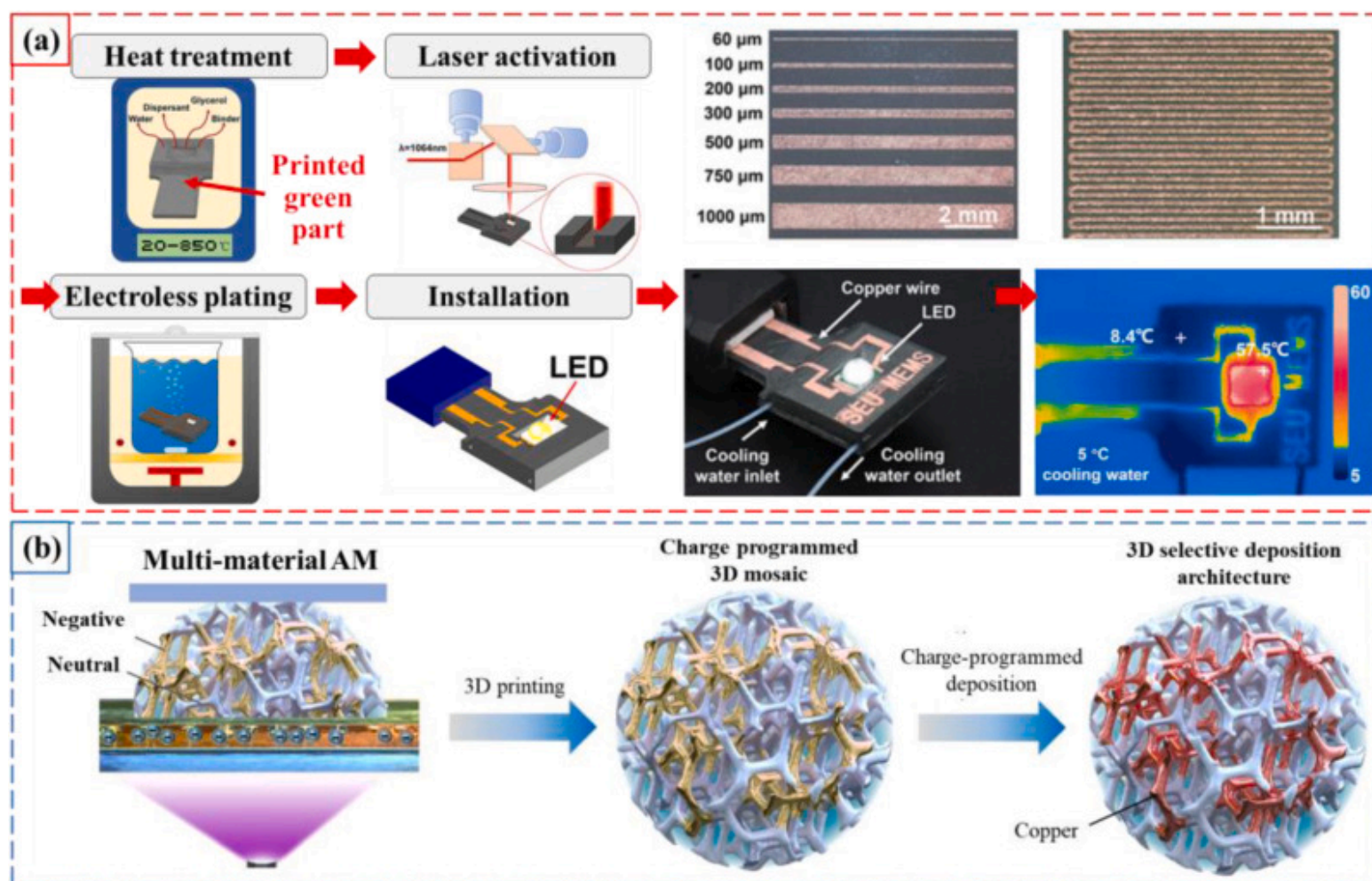
Ceramic-metal	Properties	Ref.
TiO ₂ /Ag– Ti6Al4V	Antibacterial properties	[184]
BaTiO ₃ –Ti6Al4V	Piezoelectric properties: average d_{33} of 4.92 ± 0.75 pC/N in positive electrode side, -4.68 ± 0.56 pC/N in negative electrode side; Nanoindentation hardness of 5.9 ± 0.9 GPa for Ti6Al4V, 4.0 ± 0.6 GPa for BaTiO ₃ ; Nanoindentation Young's modulus of 130 ± 14 GPa for Ti6Al4V, 106 ± 10 GPa for BaTiO ₃	[185]

The fabrication of CM-IPC is relatively straightforward compared to high-energy and low-energy input AM technologies. However, it faces limitations such as the inability to create structures with closed internal features and the problem of sufficient infiltration. Kota et al. [179] discussed various melt infiltration techniques. Mechanical pressure, centrifugal force-driven infiltration, ultrasonic-assisted processes, vacuum-assisted processes, and electromagnetic field manipulation are pivotal factors to efficiently and completely impregnating molten materials into porous scaffolds [161,187,188]. Structural characteristics like lattice type and size affect the infiltration rate [189] and optimizing the composition of ceramic or metal materials can enhance melt penetration [190], albeit with the possibility of introducing extraneous components.

2.4.3. Hybrid of AM and selective electrodeposition

Electrodeposition offers a powerful route to in-situ metallization, and its coupling with AM enables the fabrication of CMMMS [191]. Laser-induced selective metallization (LISM), which includes laser activation and metal plating processes, is a technique for creating 2D or 3D conductive pathways on intricate surfaces. During the metal plating process, metal patterns are selectively deposited onto activated locations containing activation catalysts. LISM can achieve line widths as fine as $55 \mu\text{m}$ and is widely used in manufacturing smartphone antennas [[192], [193], [194]]. The high maturity of LISM in metal plating on ceramics makes it a promising method for hybrid AM processes, especially for CMMMS [195]. Wang et al. [29] demonstrated this by 3D printing low-temperature co-fired ceramic substrates, which were then laser activated and electroplated with copper after sintering at 850°C , as shown in Fig. 14(a). The activator was mixed in the ceramic printing slurry. A challenge in this process is the compatibility of the laser activation catalyst with high-temperature heat treatment; chromium black was used as a catalyst due to its high-temperature resistance, enabling the production of high-conductivity 3D LED circuit boards with significantly reduced

operating temperatures. The conductivity of the conductive copper circuit is $5.5 \times 10^7 \text{ S/m}$, close to that of pure copper ($5.8 \times 10^7 \text{ S/m}$).



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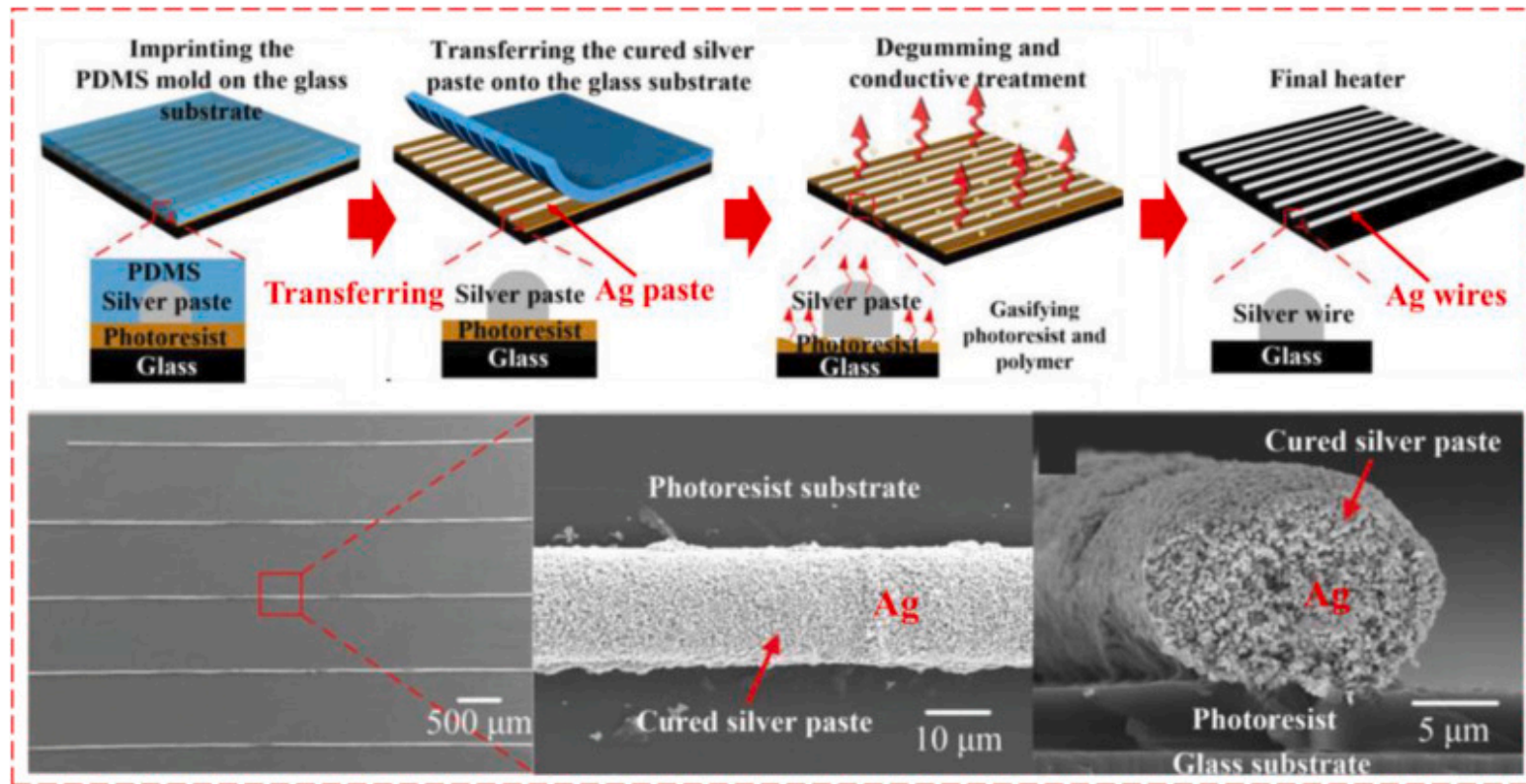
Fig. 14. (a) Basic process flow of the hybrid of AM and LISM and fabricated CMMMS device [29]. (b) Charge-programmed deposition AM method [30].

The limitation of the study is the deposition of conductive metals only on the surface of printed parts, not within internal structures. However, it is possible to create embedded metal structures through a repetitive process of printing, sintering, activation, electroplating, and further printing. Song et al. [196] proposed a method combining multi-material VP and LISM for creating complex hollow metal-plastic structures, a technique that could potentially be applied to CMMMS fabrication.

A notable advance is the charge-programmed deposition AM strategy introduced by Wang et al. [197], which delivers high-precision 3D electronics. In this approach, multi-material VP is first employed to print neutral and negative resins; subsequent electroplating then selectively deposits metallic layers only on the negative resin regions, thereby controlling the material distribution, as shown in Fig. 14(b). The authors further fabricate a complex transmit-array antenna composed of interconnected multilayer dielectric/conductive unit cells, demonstrating the unparalleled flexibility of this approach in antenna design and fabrication [30]. The prepared transmit-array antenna achieves a 94% mass reduction compared with conventional antenna configurations. Similarly, Song et al. [198] fabricated ceramic components with inert and Pd^{2+} activated domains using multi-material vat photopolymerization. Ni^{2+} , Cu^{2+} , or Ag^{+} ions were selectively deposited onto the activated domains via electroless plating. The authors further demonstrated that the resulting ceramic-metal interfaces maintained a continuous structure and stable functionality under various environmental conditions. Continued exploration of this approach holds exceptional promise for fabricating intricate CMMMS systems.

2.4.4. Hybrid of AM and transfer printing

Transfer printing technology facilitates the transfer of patterns from an intermediate carrier to various substrates including ceramics [199]. It's extensively used in photonics and electronic devices like solar cells and volumetric antennas [[200], [201], [202]]. Combined with AM, this method enables the transfer of complex 3D printed patterns onto 3D substrates. Zhu et al. [31] introduced an electric-field-driven microscale 3D printing process followed by UV-assisted transfer, allowing for the high-resolution transfer of silver paste from a mold to a substrate, which was then sintered, as shown in Fig. 15. The use of AM in the mold creation provided the transferred silver mesh with a high aspect ratio. Qi et al. [199] developed a graphene-enhanced transfer printing technique allowing metal patterns with nanoscale features to be transferred onto sapphire and SiO_2/Si wafer substrates. In addition, certain transfer printing technique showed the capability to transfer printed and sintered metal patterns onto 3D printed substrate [203]. This is another potential strategy to address the mismatch of sintering temperatures between ceramics and metals.



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Fig. 15. Schematic of the hybrid of AM and transfer printing, and the transferred Ag wires [31].

However, the bonding strength between the transferred patterns and the substrate might be limited. Transfer printing, as a low-energy integration method, is applicable to various substrate materials and minimizes substrate surface damage. With AM, the geometric shapes of printed patterns, intermediate carriers, and substrates can be customized for precise matching, allowing complex patterns to be transferred onto intricate surfaces. Most transfer printing techniques are currently applied to glass or polymer substrates [204,205], but these methods may also be relevant for ceramic substrates, warranting further research in this area.

3. Applications and AM-enabled functionalities

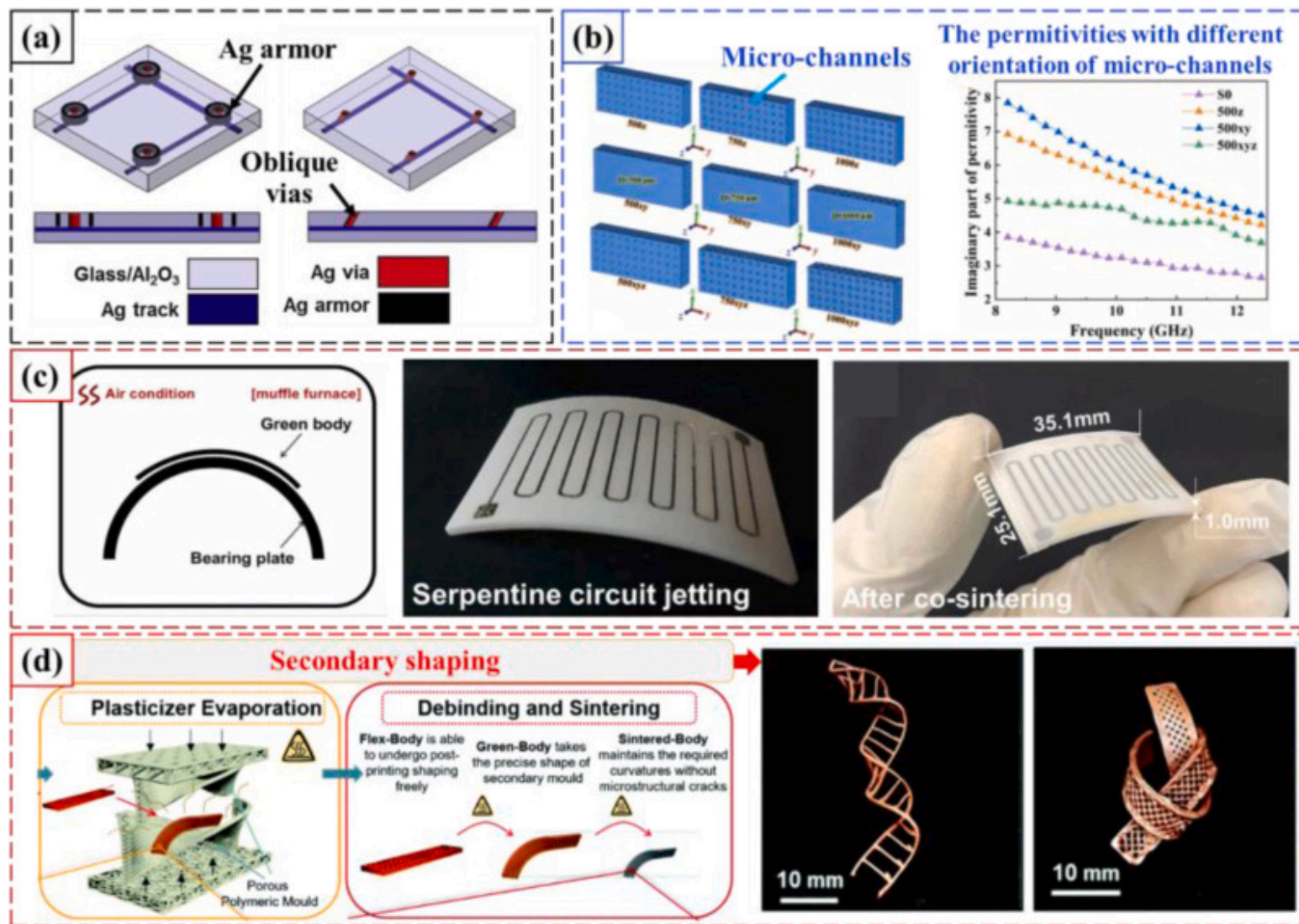
The systematic review of the diverse AM technologies suggests that these technologies are still in the developmental stage. Based on the literature survey, electronic devices represent the more mature application cases for AM-CMMMS to date, and thus are the primary focus of this section. Considerable efforts have also been directed toward enhancing mechanical properties, thermal functionalities, and biological performance of printed CMMMS, although these applications continue to face the technical challenges discussed above. Notably, this section highlights how AM offers unique advantages over conventional manufacturing routes, providing insights of the significance and potential of AM-CMMMS.

3.1. Integrated electronic systems

Multi-component/device integration can be achieved via AM technologies, further benefiting the fabrication of integrated electronic systems. This is primarily associated with low-energy input AM and hybrid AM technologies. AM can revolutionize the design and production of electronic systems in several ways discussed below.

3.1.1. Enhanced geometric complexity of printed components

Effective structural design in AM not only enhances electrical performance but also reduces device weight, which is crucial for commercial applications [206]. AM allows for the fabrication of intricate structural features like microstrip lines and vias that are unattainable with conventional methods. Raynaud et al. [24] demonstrated the creation of complex geometric structures and metallic networks for CMMMS electronic devices, including oblique vias and armored vias (Fig. 16(a)).



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Fig. 16. The capability of AM in enhancing component geometric complexity. (a) Multilayer co-fired ceramics with intricate 3D structures [24]. (b) Performance variations with structure design control [207]. (c) Conformal printing on curved surfaces [146]. (d) AM-produced 3D structures through secondary shaping [125].

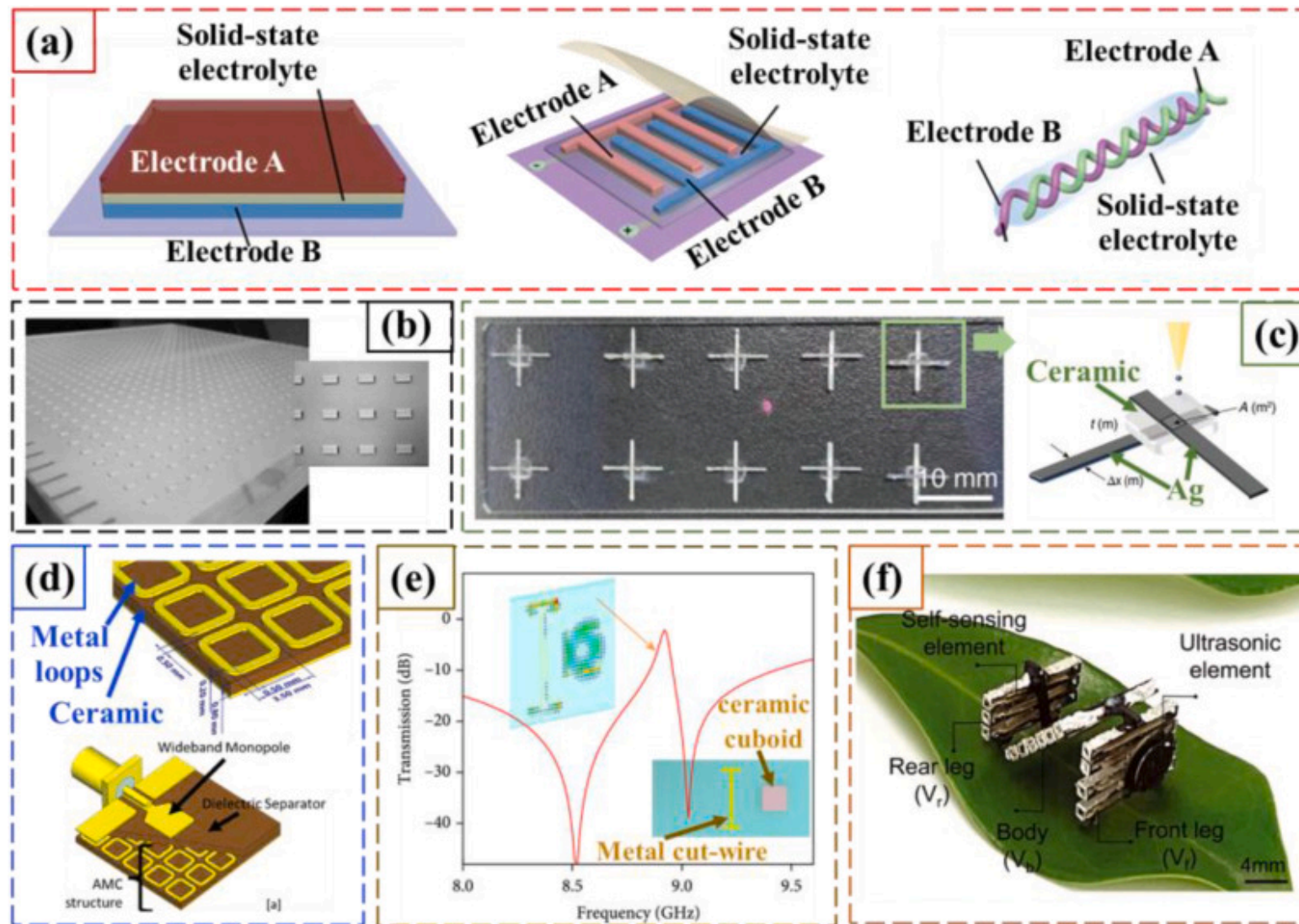
Controlled structure design can also alter performance characteristics. For example, adjusting structural parameters like size, porosity, and micro-channels can control the directional electromagnetic properties of materials, such as dielectric constant and permeability [207,208], as shown in Fig. 16(b). A zirconia flat grin lens with a variable dielectric constant from 3 to 23 was achieved by altering filling ratios [208]. The Luneburg Lens, composed of 7494 different-sized unit cells, showcased the performance gradient due to structural variations, enabling electromagnetic transitions through material interfaces and minimizing reflections or sculpting radio frequency waves [46,209,210]. These capabilities significantly improve electronic device performance.

Conformal printing on non-planar surfaces is another key advantage of AM, simplifying the production of uniquely shaped CMMMS electronic components. Traditional components are mostly planar, whereas AM allows for in situ conformal printing to create devices tailored to the shape of the carrier [211,212]. Liang et al. [146] integrated a curved LTCC device with a serpentine conductor pattern, maintaining the original shape after co-firing at 900°C by controlling the similar solids content of the metal conductor (Ag) and the substrate ($\text{SiO}_2/\text{H}_3\text{BO}_3$) (Fig. 16(c)). Curved LTCC can be achieved by placing the printed green part on an alumina mold with the desired curvature during sintering, leveraging the deformation behavior of the green part during heat treatment for secondary shaping [213]. Zhang et al. [125] prepared complex ceramic and metal structures using this approach, further augmenting the structural complexity of AM-CMMMS through secondary forming (Fig. 16(d)). When combined with the aforementioned conformal printing technology of CMMMS [146], it is anticipated that the structural complexity of AM-CMMMS can be further augmented through secondary forming.

3.1.2. Multi-component/device integration

- a. *Optimized components layout in a device:* The precise positioning and interconnections of components significantly influence device performance. For instance, the layout of metal wires on a ceramic insulating surface has been found to impact the surface flashover of electronic devices, which is widely acknowledged as a primary factor influencing system insulation failures [5]. AM allows for the creation of complex geometric structures on ceramic substrates, such as honeycomb and tubular electrodes, optimizing electrode layout and density [144]. This precision is crucial in applications like electronic devices, where wire layout affects surface flashover and system insulation. Miniaturized electrodes on LTCC substrates, for instance, ensure effective connections with larger contact areas [218]. In battery technology, including solid-state and solid oxide fuel

cells, the arrangement of metal electrodes and ceramic electrolytes enhances performance (Fig. 17(a)) [214,219,220].



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Fig. 17. The role of AM in optimizing component layout and developing multiphase metamaterials. (a) Intricate spatial arrangements of printed components [214]. (b-c) Individually designed components in large-scale manufacturing [215,216]. (d-e) CMMMS Metamaterials [4,217]. (f) 3D printed mobile robot with active sense and move functions [10].

AM also enables cost-effective increases in component surface area, volume, layer count, and array quantity, boosting device performance. It allows for easy modification of electrode overlap and dielectric layer thickness to alter capacitance values in capacitors. Five Ag/BaTiO₃/Ag parallel-plate capacitors printed by IJP were stacked into multilayer components in order to achieve capacitor density several times that of traditional parallel-plate capacitors [221]. Fig. 17(b) displays various capacitors with different capacitance values printed on a dielectric substrate [215]. These capacitors can be designed in parallel for higher overall capacitance. Dossou-Yovo et al. [216] printed 784 multilayer ceramic capacitor (MLCC) components, each with 15 Ag electrodes and 14 MgTiO₃ dielectric layers, demonstrating the potential for creating high-density capacitor arrays (Fig. 17(c)).

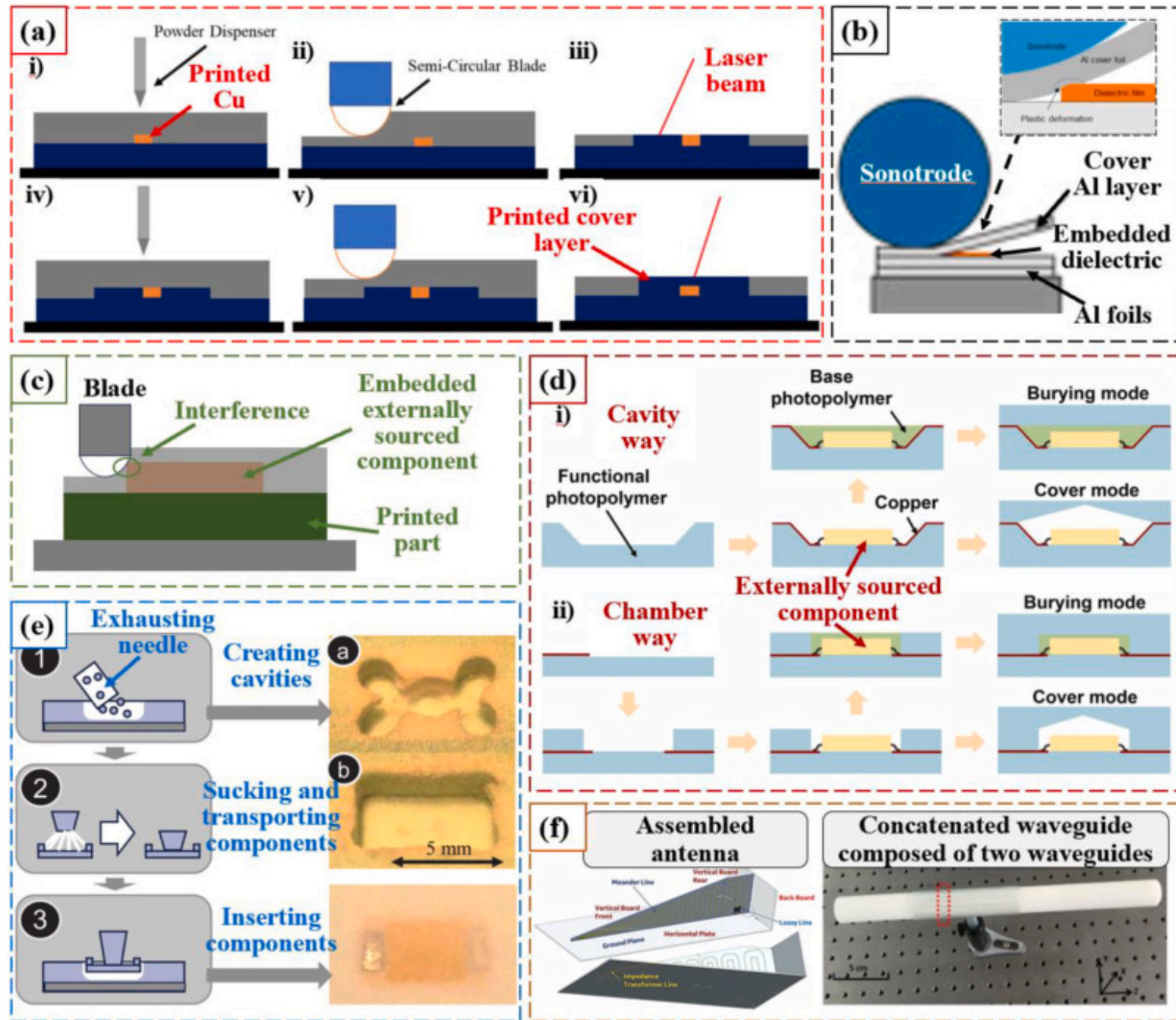
- b. *Development of multiphase metamaterials:* Unique structural designs facilitated by AM have promoted the creation of metamaterials that surpass conventional material properties. In electronic devices, conductors (metals) are often separated by dielectric materials (ceramics) [[222], [223], [224]]. The spatial arrangement of ceramics and metals in electronic devices can produce various metamaterials, including electromagnetic, wave-absorbing, and optical metamaterials [[225], [226], [227], [228], [229]]. Goulas et al. [4] demonstrated a fully functional multilayer ceramic-metal antenna structure with a metamaterial artificial magnetic conductor (Fig. 17(d)), highlighting the role of AM in metamaterial fabrication. Fan et al. [217] presented a ceramic-metal hybrid metamaterial that, due to electromagnetic coupling, offers distinct nonlinear tunability, broadening the operating band and enhancing light-matter interactions (Fig. 17(e)). The implementation of such metamaterial designs through AM is expected to significantly elevate device performance.
- c. *AM of integrated systems:* The fabrication of integrated systems through AM enhances performance and showcases the unique capabilities of AM. For instance, antennas with controlled material distribution in 3D space, can achieve high gain, radiation efficiency, and operational bandwidth [230]. An integrated lens antenna system, combining a dielectric lens and a microstrip patch antenna, significantly increases antenna

gain [231,232]. Phased array antenna systems, crucial in monitoring and communication, benefit from 3D-printed antenna units and phase shifters. Tunable microwave devices, formed by combining printed capacitors, have transformed the design and manufacture of such devices [233]. A fully 3D-printed tunable microwave subsystem consisting of 26 circuit elements demonstrated excellent radiation performance [234]. Cui et al. [10] successfully utilized AM to build complex shapes for a unified miniaturized mobile robot with actuation and sensing functions (Fig. 17(f)). The creation of integrated systems is made possible by leveraging distinct performance characteristics of both ceramics and metals.

In essence, AM enables customizable electrical integration across various components/devices, prompting a paradigm shift in electronic device design. Traditional multilayer co-fired ceramics manufacturing, involving steps like drilling and laminating, is limited in creating complex structures and lateral shielding [3]. Integrated AM challenges the “multilayer” concept, allowing metal patterns to span across layers and substrates to adopt unconventional shapes. This shift may lead to significant changes in the design and production processes of various electronic devices and systems.

3.1.3. Methods of integrating multiple components during printing process

Achieving end-use devices through AM requires the integration and packaging of multiple components during the printing process. This shift towards non-assembled manufacturing aims to produce fully functional end-use devices directly from the printing process. As shown in Fig. 18(a), complex conductive networks, vias, connecting pads, and passive components can be integrated within AM or hybrid AM processes [[1], [2], [3],235,236]. Embedding externally sourced components during printing presents a challenge, requiring strategies to integrate them effectively, as discussed below.



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Fig. 18. Methods for integration and packaging of electronic components for AM-CMMMS. (a) Basic process flow of printed components [1]. (b) Using ultrasound to integrate components into printed matrix [238]. (c) Interference of embedded externally sourced component on the motion of blade. (d-e) Method for embedding externally sourced components into cavities [239,240]. (f) Extending the complexity of components by assembling several printed parts [25,241].

Ultrasound-assisted AM can integrate metal circuits and components into solid substrates [237,238], as shown in Fig. 18(b), but high energy input may damage the printed parts and components. A gentler approach is to create receptacles for components, which can be achieved by fabricating cavities in the printed parts. However, the protrusions resulting from embedded components may interfere with the motion of the scraper, impacting material spreading and printing, as shown in Fig. 18(c). Wang et al. [239] proposed strategies to embed electronic components like chips and resistors into these cavities, pausing the printing process, and then resuming printing after component installation, as shown in Fig. 18(d). For PBF processes, cavities can be formed using an exhausting needle to place components [240], as shown in Fig. 18(e). Components can be encapsulated with a curable resin for further printing on a flat surface, with the potential to replace the resin with ceramic or other materials. Surface-exposed components can be sealed directly with resin [218]. It is important to note that embedded components and encapsulation resins must withstand high temperatures during heat treatment.

While non-assembled methods are appealing, integrating multiple 3D printed modules through assembly and joining is also a viable approach [25,241], as shown in Fig. 18(f). Some simple post-processing steps may be necessary to achieve fully functional end-use devices.

3.2. Non-electronic applications with customizable functionalities

For non-electronic applications, AM enables the realization of customizable functions through the efficient integration of ceramics and metals, the flexibility of process parameters, and multiscale structural designs.

Integrating ceramic and metal materials can achieve designed functions of the printed parts. The typical example is the unique combinations of antagonistic mechanical properties, such as in gears where metal bodies are coated with wear-resistant ceramics at high-wear locations [40], and in armor where the hardness from outer ceramic region resists penetration while the ductility from inner metal region prevents secondary damage [242,243]. The ability to achieve such multi-material integration largely

depends on the configuration of the printing equipment. A critical consideration is preventing cross-contamination between different materials while ensuring recyclability. Typically, materials are supplied through multiple nozzles or hoppers [1,57,244]. The supplied material can be spread onto the powder bed with a blade or deposited at specified locations by nozzles. Alternative feeding strategies include multi-material AM manipulators [245] and hybrid multiple feeding methods [65,246]. It is worth noting that irregular ceramic powders, such as SiC, exhibit poor flowability and are prone to agglomeration, highlighting the need for improved powder delivery strategies. Wei et al. [247] proposed a hybrid ultrasonic and motor vibration integrated dispensing method, which enabled a smooth, continuous flow rate of irregularly shaped SiC powders and SiC-316L composite powders.

In addition, through flexible process adjustments, AM can adjust the material properties in real-time from the micro (such as the melt pool) to the macro scale. For instance, changing the laser power and scanning strategy significantly alters the microstructures, and adjusting the atmosphere can change the material composition [43,76]. These changes further affect mechanical properties, electromagnetic properties, etc., thereby customizing the material properties at different regions of the printed component. In the biomedical sector, AM allows for the customization of implant properties by adjusting parameters like laser power and gas concentration [69], enhancing their suitability for specific medical applications.

Furthermore, AM enables the precise manufacturing of complex structures across multiple scales. For example, in armor manufacturing, AM allows for macroscopic laminated and gradient structures that harness the combined advantages of both materials to enhance armor performance [248]. This precision control over layer shape and size directly affects the protective performance of armor [249], which can be further endowed with multifunctional properties like optimized ballistic resistance, thermal protection, and electromagnetic shielding [250,251], potentially revolutionizing military tactics [252]. Notably, metamaterials leverage specialized structural arrangements to exceed the properties of conventional materials. AM has contributed to the development of customized metamaterials for a variety of applications [[253], [254], [255]]. Multiphase metamaterials, in particular, offer expanded design possibilities for complex functionalities beyond those of single-phase counterparts [47].

Surface microscale structural design also enables customized functionalities of printed parts. AM enables diverse functional designs, including mechanical properties control [256], enhanced bioactivity [257], acoustic tuning [258], electromagnetic performance control [207] and even friction law customization [259]. These examples demonstrate the capability of AM to achieve multi-function integration with customized performance metrics for each function, underscoring the importance of high

freedom degree in structural design provided by AM for creating special functions in CMMMS [260]. The cross-scale manufacturing potential of AM is expected to integrate multiple functions effectively, warranting more investigation in the future.

4. Summary and future trends

4.1. Summary

To gain a better understanding of the relationship between CMMMS and AM, CMMMS can be conceptualized as comprising three functional components: a ceramic phase, a metal phase, and the integrated CMMMS itself. Single-phase material parts are capable of leveraging the inherent properties and phase structures to deliver specific functionalities. At the same time, CMMMS give rise to integration of multiple functionalities and devices. The integration and coupling of these multiple devices significantly enhance the performance of CMMMS, leveraging their multifunctional interactions. In this context, AM plays a pivotal role in providing robust manufacturing capabilities that support the production of CMMMS, as well as enhancing individual functions and facilitating the coupling of multiple functions.

This review comprehensively examines AM-CMMMS technologies, each with unique traits, capabilities, limitations, and future trajectories. High-energy input AM often leads to internal defects and is still in its early stages, and current efforts are concentrated on improving interfacial bonding and mechanical performance. Low-energy input AM, with a gentler process, is extensively researched for electronic device manufacturing. Hybrid AM combining with conventional manufacturing provides a feasible approach to produce high-performance end-use devices. We also highlight two significant AM contributions—manufacturing of integrated functional systems and customizable multi-function integration—that are poised to shift the paradigm of device design and production.

Despite these advances, several critical limitations remain. First, a fundamental understanding of interface evolution mechanisms—including the effects of printing parameters on elemental diffusion, reaction layers, residual stress accumulation, and defect generation—is still incomplete, limiting the ability to predict and control interfacial performance. Second, comprehensive evaluations of interfacial bonding quality and long-term reliability are still lacking. Addressing these challenges requires continued research efforts in materials science, process engineering, and characterization methodologies to unlock the full potential of AM-CMMMS.

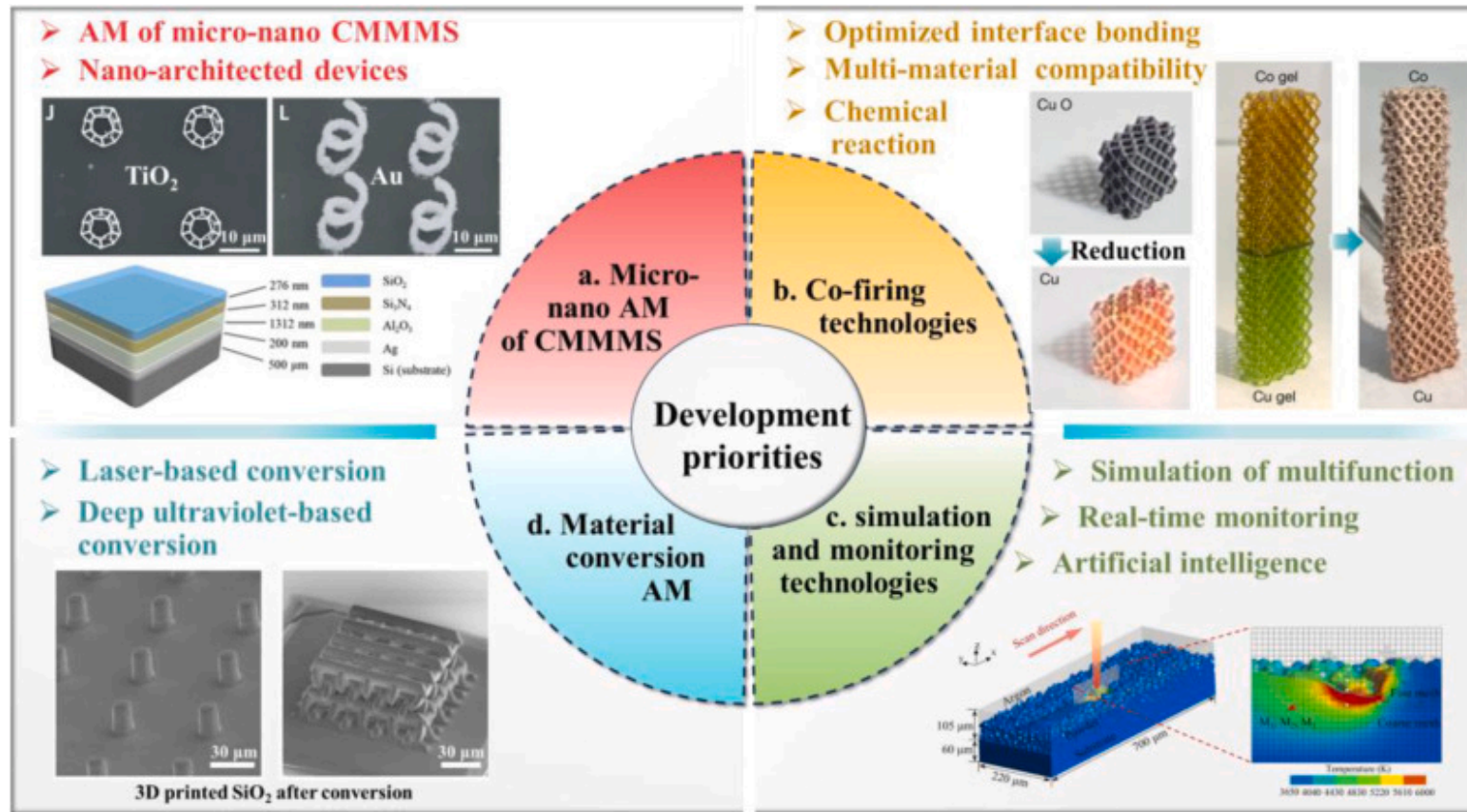
4.2. Future trends

Building on the comprehensive review of various technologies and applications discussed above, the current short-term challenges can be categorized into two major issues. The first is to further explore breakthroughs in the functionalities and applications of AM-CMMMS. The second is to improve interfacial bonding quality while maintaining the overall performance of printed parts. Based on this, this section outlines several key priorities. Micro-nano scale AM-CMMMS is expected to unlock new properties and potential applications, serving as a promising entry point for the first issue. For the second issue, efforts can be focused on co-firing, simulation and monitoring, and material conversion AM technologies. [Fig. 19](#) offers a visual summary of these pivotal research directions, which are detailed point by point below.

- a. *Micro-nano AM of CMMMS*: The nanoscale size effect is pivotal for achieving unique properties in printed parts, especially for high-resolution devices like solar cells and chips. In solar technology, metals (such as Au, Ag, Al, W, Ti, Zr) and ceramics (such as TiN, TiO₂, SiO₂, Si₃N₄, Al₂O₃, ZrO₂, ZnO) are crucial for solar absorption or reflection [[266](#), [267](#), [268](#), [269](#), [270](#), [271](#), [272](#)]. Nano-laminated structures of ceramics and metals have shown exceptional performance in solar applications, highlighting their importance for solar device advancement [[273](#), [274](#), [275](#)]. Micro-nano AM promises enhanced performance through intricate structural designs, such as tuning absorption peaks by adjusting metal disc diameters and spacer layer thicknesses [[276](#), [277](#), [278](#)]. Advances in micro-nano AM have been significantly expanded, basing on the principles of light [[261](#),[279](#)], electricity [[280](#), [281](#), [282](#), [283](#)] and chemistry [[284](#),[285](#)]. It is worth paying particular attention to methods applicable to various materials [[261](#),[285](#)]. A high-throughput 3D printing technique introduces compositional gradients at the microscale, enhancing the ability of AM in controlling multi-material distribution [[286](#)].
- b. *Co-firing technologies of CMMMS*: Co-firing has received insufficient attention in previous work. Expanding the range of printable materials is essential for enhancing the functionality of multi-material structures. While low-energy input AM techniques show higher maturity in this regard, further optimization of co-firing technologies for CMMMS is needed to ensure material performance and overall functionality. Innovations in heating and sintering technologies are a promising direction. The co-firing process also allows for secondary shaping and material transformation through chemical reactions, transforming multi-metal precursor

scaffolds into metal multi-material structures through sintering and reduction post-treatment [263]. This approach can be extended to ceramic precursors [287] for the preparation of CMMMS.

- c. *Simulation and monitoring technologies*: With the increasing complexity of multi-parameter coupling, it is essential to employ advanced simulation techniques [57,119,288] for the prediction of both printing process and final performance outcomes. Simulation technologies should enable the modeling of multifunctionality within CMMMS, thereby assisting in structural optimization, directional design, and modular functional design. Improving printing qualities through process optimization hinges on closed-loop control, making in-situ monitoring important. Establishing advanced real-time monitoring methods to build a process database, and integrate it into manufacturing through artificial intelligence will be highly beneficial. Combining such database with simulations, particularly through digital twin technologies, holds promise for revealing underlying mechanisms and enhancing process control capabilities.
- d. *Material conversion AM*: Material conversion enables the processing of materials that are otherwise hard to print directly, such as ceramics, and potentially mitigate significant property mismatches at the interface. Section 2.2.3 reviews several related techniques, including laser-based routes that can be readily integrated into existing AM systems. Li et al. [265] demonstrated a mild photochemical route in which intricate 3D-printed resin structures are converted to silica glass by deep ultraviolet exposure in an ozone atmosphere. This work highlights the disruptive potential of material conversion AM.



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Fig. 19. The perspectives in future R&D of AM-CMMMS. Images source from Refs. [[261], [262], [263], [264], [265]].

CRedit authorship contribution statement

Yongqi Chen: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Changhui Song:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Jiaqi Chen:** Writing – review & editing, Visualization. **Yongqiang Yang:** Supervision, Resources. **Xiaobin Hong:** Writing – review & editing. These authors contributed equally to this work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The following is the Supplementary data to this article:

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Multimedia component 1.

[Recommended articles](#)

Data availability

No data was used for the research described in the article.

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